Licensing

Use a **VCF Operations** instance and the **VCF Business Services** console as a VI administrator to manage and assign licenses.

In This Chapter

- Licensing Overview
- Registering VCF Operations and a License Server with the VCF Business Services Console
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Getting to Know VCF

To apply this documentation, you must be acquainted with the VCF Product Overview, VCF Design, and VCF Release Notes.

SDKs, APIs, and CLI for VCF Administration

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See Administration SDKs, APIs, and CLI.

Licensing Overview

You license your VMware Cloud Foundation (VCF) fleet or VMware vSphere Foundation (vSphere Foundation) environment by using theVCF Operations instance that manages them and the VMware Cloud Foundation Business Services console (vcf.broadcom.com).

What's New in 9.1

- Before you can license your assets, you must register VCF Operations and a license server with the VCF Business Services console, and you must add licenses to the license server. For more information, see Registering VCF Operations and a License Server with the VCF Business Services Console.
- You can assign a license directly to specific assets other than vCenter instances, such as ESX hosts, and vSAN clusters. The licenses that you assign directly to these assets override the vCenter license that is assigned automatically to them. For more information, see What is an Override License.

Overview of Core Procedures and Concepts in 9.1

Subscriptions

To license your environment, you purchase a term-based subscription for a product with a specified capacity. For more details, see Licensing Model.

Licenses

A license is an object that entitles you to use the products that you purchased subscriptions for. A license can span multiple subscriptions or, when you split the default license, a license can be a portion of a subscription.

Licenses and product components interaction

VMware Cloud Foundation and vSphere Foundation include different sets of product components. You purchase a subscription for a specific product. With the provided license, you license all product components by using a registered VCF Operations instance.

- With your subscription for a product, you receive both a version 9+ default license, and license keys for version 8.x of the product. You can decide which version of the product to deploy and license. Your total used capacity must not exceed your purchased capacity.
- After you assign a license to a vCenter instances, the other product components added to the licensed vCenter instances are automatically licensed. For more information, see Assign a Primary License to a vCenter Instance.
- One VCF Operations instance can manage a mix of different license types and licenses for different products. For example, you can manage both VCF and VMware vSphere Foundation primary licenses in the same VCF Operations instance, and add-on licenses for different product components.

Default licenses

The initial license that is automatically created in VCF Business Services console after you purchase a subscription is called a default license. The default license is a pool of capacity from all active subscriptions for the same product in the same Site ID. Available capacity is displayed in the unit of measure for the specific product.

For example, if you purchase a VCF subscription for 500 cores, you receive a default license for 500 cores of VCF and you also receive a default license of 500 TiBs of vSAN, that are added in a separate license. If, on top of that, you purchase a vSAN add-on subscription for 200 TiBs, the new number of TiBs is added to your default vSAN license, and its capacity becomes 700 TiBs.

> **Note:**
>
> For more information about using your Broadcom Site ID, see Broadcom knowledge base article 142873.

Types of licenses

There are two types of licenses - primary licenses, such as VMware Cloud Foundation and VMware vSphere Foundation licenses, and add-on licenses, such as vSAN add-on capacity or VMware Private AI Foundation with NVIDIA licenses. You have a single license capacity provided per product.

For example, for VMware Cloud Foundation, the single license capacity provided for that product is *VMware Cloud Foundation (cores)*. This is a primary license and you must assign it to your environment before you can assign an add-on license. Components are licensed automatically after you assign the primary license to a vCenter instance. With VCF and vSphere Foundation, you also receive *VMware vSAN (TiB)*capacity. You also receive *VMware vSAN (TiB)* capacity when you purchase vSAN as an add-on. The vSAN TiB license is called an add-on license.

For more information about types of licenses, see Licensing Model.

Supported add-on licenses

In 9.1, the two supported add-on licenses that you can manage in the VCF Business Services console are VMware vSAN (TiB) and VMware Private AI Foundation with NVIDIA (cores).

You can purchase subscriptions for add-ons which are compatible with your products. For information about available add-ons for each product, contact your sales team. Add-ons might not follow the same licensing model as VCF and vSphere Foundation. You can find information about how to license specific add-ons within their respective documentation.

VCF Operations registration modes

You can register VCF Operations in the VCF Business Services console either in connected or in disconnected mode. Connected mode is recommended because it simplifies license management.

License usage reports

License usage reports are required at least once every 180 days to maintain your licenses. You must update your licenses to confirm that the license usage report was submitted.

- If VCF Operations is registered in connected mode, license usage data is sent to the VCF Business Services console automatically, and licenses can be updated either automatically or with a click of a button. See Updating Licenses and Viewing the License Usage File.

- If VCF Operations is registered in disconnected mode, to report license usage, you must generate a usage file and upload it in the VCF Business Services console (vcf.broadcom.com). See Updating Licenses and Viewing the License Usage File.

Subscription expiration

The total capacity of a license at any point in time equals the sum of the license capacity from all contributing subscriptions that are active. When a subscription expires, its contribution to each license is removed. If other subscriptions still contribute capacity, the license remains active but with reduced capacity. When the last contributing subscription expires the license expires.

License expiration

License expiration applies when the last contributing subscription expires or if usage is not reported and the license is not refreshed.

- Notifications are displayed before the license expiration.
- After the license expires, you have 90 days to update your license. License assignments are impacted during this 90 day period and notifications appear in most components, including VCF Operations and vCenter.
- 90 days after the expiration, your environment is impacted in several ways, including:
  - Management operations of various components might be prevented;
  - ESX hosts disconnect from vCenter, most host management operations are prevented;
  - You cannot start workloads.

    Note:

    Existing workloads are not proactively stopped.
- You can recover expired environments by downloading an updated license file.

Usage analytics

You can check your license consumption analytics in your VCF Operations instance, and the VCF Business Services console. For more information about usage analytics, see License Usage Analytics.

Licensing Terminology

As part of the licensing workflows, specific terminology is used to refer to different licensing-related objects and actions.

Term	Related nouns	Related actions	Description
Subscription			You purchase a subscription for a specific product. You decide what capacity to purchase for the subscription. Subscriptions are term-based.
Assets			Any object in your environment that requires licensing. Assets are vCenter instances, ESX hosts, VCF Operations instances, NSX, and others. You assign a license to a vCenter instance, and all other assets, which are connected to that vCenter instance, are licensed automatically. You can also assign a license directly to assets which use license capacity such as an ESX host, a vSAN cluster, a VMware Private AI Foundation with NVIDIA cluster, or a DRS cluster. The license that you assign directly overrides the license assigned through the vCenter instance to which the asset is added.
License			An object that entitles you to use the products that you purchased subscriptions for. A license can span multiple subscriptions or, when you split the default license, a license can be a portion of your total subscription capacity.
		Create/Split	Performed in the VCF Business Services console. The process of generating a new license. You allocate to it a portion from the capacity of a default license. For more information, see Split Default Licenses.
		Add (to VCF Operations and license server)	Performed in the VCF Business Services console. The process of associating the license with a license server. Then you can assign that license to assets from the VCF Operations instance. For more information, see Add a License to VCF Operations and License Server.

		Remove (from VCF Operations and license server)	Performed in theVCF Business Services console. When you remove a license from a license server, the license is no longer available to assign in the VCF Operations instance, and you can add the license to a different license server and, respectively, a different VCF Operations instance. For more information, see Remove a License from a License Server and the Respective VCF Operations Instance.
		Delete	Performed in the VCF Business Services console. After you delete a license, it is no longer available and the allocated capacity from this license is automatically merged into the capacity of the corresponding default license. For more information, see Delete a License.
		Assign	Performed in VCF Operations. The action of entitling a vCenter instance the cores or TiBs on assets within the vCenter instance, including hosts or vSAN clusters, and any connected component, for example NSX. For more information, see Assign a Primary License to a vCenter Instance or Assign an Add-on License to a vCenter Instance. You can also assign a license directly to assets which use license capacity such as an ESX host, a vSAN cluster, a VMware Private AI Foundation with NVIDIA cluster, or a DRS cluster. The license that you assign directly overrides the license assigned through the vCenter instance to which the asset is added. For more information, see What is an Override License.
	Default license		An initial license that is automatically created in the VCF Business Services console after you purchase a subscription. A default license is a pool of capacity from all active subscriptions for the same product or product component in the same site. Depending on your subscription, you receive a set of default licenses. For example, with a vSphere Foundation subscription you receive a vSphere Foundation default license, and a vSAN default license.
	Split license		A license that you create manually. You allocate to it a portion of the capacity of a default license. For more information, see Split Default Licenses.
	Primary license		VMware Cloud Foundation (cores) and VMware vSphere Foundation (cores) licenses.
	Add-on license		You purchase them separately from the primary license, or, for some product components such as vSAN, you also receive capacity for the add-on license with your primary product purchase. You can assign add-on licenses to vCenter instances only after you assign a primary license. For example, in VCF 9.1, two of the supported add-on licenses are *VMware vSAN (TiB)* and *VMware Private AI Foundation with NVIDIA (cores)*.
	Override license		An override license is a license that you assign directly to assets connected to a vCenter instance. An override license can be a primary or an add-on license, it could be the default license or a split license.
	vCenter license		The license that you assign to a vCenter instance. You can assign both primary and add-on licenses to a vCenter instance. When you assign a license to a vCenter instance, it is automatically assigned to all assets connected to that vCenter instance that correspond to the type of license. For example, a VCF license is assigned to the ESX hosts, and a vSAN license is assigned to the vSAN clusters. If override licenses are assigned to the assets, the vCenter license is not assigned to them.
Evaluation mode			After you install a product or component, it operates in evaluation mode for up to 90 days. During that period, you must license your environment. Evaluation mode ends when a license is assigned or at the end of the 90-day period, whichever comes first.

License category			The license categories are default, split, and shared.
License type			The types of licenses are primary and add-on.
License server			A license server is added to a VCF Operations instance to store the licenses you want to assign from this VCF Operations instance. You must add at least one license server to each VCF Operations instance that you use for license management.
License unit			The unit of measure for capacity, for example, cores or TiBs.
Capacity			The number of cores or vSAN storage TiBs.
		Allocate	The process of adding capacity to a license.
Allocated capacity			A portion of the default license capacity that you added to a split license. Allocated capacity can be measured in cores or TiBs.
		Increase	You can increase the capacity allocated to a split license at any time, and up to the available capacity in the default license. For more information, see Change License Capacity.
		Decrease	You can decrease the capacity allocated to a split license at any time. You cannot decrease it below the used capacity for this license. For more information, see Change License Capacity.
Total capacity			The total capacity of a product equals the combined capacity of the default license and all other licenses split from that default license.
Used capacity			Per license, the actual capacity used by the assets in your environment.
Unused capacity			Per license, the part of the allocated capacity that is not yet used by assets in the vCenter instances to which the license is assigned.
Licensed assets			Any object in your environment that is licensed, either automatically with a vCenter license or directly with an override license.
	Reclaim [used capacity]		If you delete an asset from VCF Operations, but the license capacity is not automatically freed up, you can reclaim that capacity manually. For more information, see Reclaim Used Capacity .
Over use			When a license has more used capacity than its currently allocated capacity.
Pooling			Combining the total capacity of subscriptions for the same product in one license.
Registration file			A file used to identify a VCF Operations instance, and list all license servers added to this VCF Operations instance. The name format of the registration file is: registration-<vcfOpsName><timestamp>.registration
Verification file			A file that verifies the integrity of a license server. You download a verification file from the VCF Business Services console and import it in VCF Operations. The name format of a verification file is: verification-<vcfOpsName><timestamp>.verification
Confirmation file			A file that completes the verification of the integrity of a license server. You download a confirmation file from VCF Operations and import it in the VCF Business Services console. The name format of a confirmation file is: confirmation-<vcfOpsName><timestamp>.confirmation
License file			The license file contains the licenses added to a VCF Operations instance from the VCF Business Services console. The name format of the license file is: license-<vcfOpsName><timestamp>.lic

License usage file			License usage files contain information on the capacity of each license used by a VCF fleet or a vSphere Foundation environment. VCF Operations creates the license usage files and sends them to the VCF Business Services console. For more information about the license usage file, see Updating Licenses and Viewing the License Usage File.

Licensing Model

The licensing model is the same for VCF and vSphere Foundation. You assign licenses to vCenter instances. The other product components that are connected to the licensed vCenter instances, are automatically licensed. Starting with version 9.1, you can also assign a license directly to assets that use license capacity, for example, ESX hosts, and vSAN clusters. The licenses that you assign directly to assets override the license that is automatically assigned.

How Capacity Pooling Works

When you purchase multiple subscriptions for the same product, capacity is pooled together in one default license per product. For example, if you purchase a subscription for 200 cores for VCF, and then you purchase another subscription for 300 cores, you have one default primary license for the product with 500 cores total capacity, instead of two licenses with 200 and 300 cores capacity. If the different subscriptions have different end dates, the number of cores is reduced on the end date of each subscription, but you can use the capacity for which you have an active subscription.

vSAN capacity, when purchased through VCF, vSphere Foundation, or a vSAN add-on, also pools together in one default add-on license. For example, when you purchase a subscription for 300 cores for VCF, and 50 TiBs as a vSAN add-on, you have one default vSAN license with 350 TiBs total capacity.

To pool together into the same default license, subscriptions must meet the following requirements:

- Must be for the same product, for example, VCF, or vSphere Foundation.
- Must have the same unit of measure, for example, cores or TiBs.
- Must have the same license (Site ID).
- Must not support overage billing. VMware Cloud Services Provider (VCSP) partner subscriptions are not eligible for pooling.

Note:

For some products, you receive multiple default licenses. For example, with a VCF subscription you receive both a default VMware Cloud Foundation license and a default vSAN Enterprise license.

What Are Primary Licenses?

Primary licenses are the licenses for VCF and vSphere Foundation. They are based on a core license unit. The only assets in your environment which use the capacity of your primary licenses are ESX hosts. To calculate the capacity you need for your environment, you need the total number of the physical CPU cores for each physical CPU on all ESX hosts in your environment. Most products have a minimum consumption rule of 16 cores per physical CPU. CPUs that have fewer than 16 cores use 16 cores. For example, if you have 1 ESX host in your inventory, with 1 CPU, and 8 CPU cores per CPU, this ESX host uses 16 cores of your license total capacity because it is the minimum license capacity.

Number of ESX Hosts	Number of CPUs per Host	Cores per CPU	Required License Capacity
1	1	8	16
2	2	8	64
2	2	16	64
2	2	24	96

For more information about calculating the license capacity you need for your environment, see Broadcom knowledge base article 95927.

You assign primary licenses to your vCenter instances and optionally as overrides to individual hosts. When a primary license is assigned to a vCenter instance, the other components of the product are licensed automatically with the license you assign to the vCenter instance.

To assign licenses to your vCenter instances, you use a VCF Operations instance. For more information about the licensing workflow, see High-Level Licensing Workflow .

Note:

The difference in the licensing model for VCF Edge and VMware Cloud Foundation is that VCF Edge has an 8 CPU cores per physical CPU minimum. For more information about VCF Edge license terms, see the Specific Program Documentation for VCF Edge.

Evaluation Mode

After you install a product or component, it operates in evaluation mode for up to 90 days. During that period, you must license your environment.

Evaluation mode applies to new deployments and upgrades from version 8.x to version 9.1. Examples of when evaluation mode applies:

1. Deployment of ESX 9.1 or upgrade from ESXi 8.x to ESX 9.1. The ESX host is in evaluation mode for up to 90 days from first boot.
2. Deployment of vCenter 9.1 or upgrade from vCenter 8.x to vCenter 9.1. The vCenter instance is in evaluation mode for up to 90 days.
3. Deployment of VCF Operations 9.1 or upgrade from Aria Operations 8.x to VCF Operations 9.1. The VCF Operations instance is in evaluation mode for up to 90 days.
4. When you add an ESX host to a licensed vCenter instance but there is no available license capacity to license the host, the ESX host remains in evaluation mode for up to 90 days. The evaluation period starts when ESX is installed, not when it is added to the vCenter instance.

Note:

ESX hosts that you provision as stateless hosts by using Auto Deploy have no evaluation period in version 9.0 and 9.1.

Evaluation mode ends when a license is assigned, or at the end of the 90 days. If a valid license is not assigned within the 90-day evaluation period, your environment is impacted in several ways, including:

1. Management operations of components are prevented.
2. ESX hosts disconnect from vCenter.
   a. If the 90-day evaluation period of a vCenter instance expires, all hosts added to the vCenter instance are disconnected from it.
   b. If the 90-day evaluation period of an ESX host expires, but the vCenter instance is licensed, only the ESX host with an expired evaluation period is disconnected from the vCenter.
3. You start workloads.

Note:

Existing workloads are not proactively stopped.

If the evaluation period expires, you can recover the operations in your environment by assigning a valid license.

1. If the 90-day evaluation period of a VCF Operations instance has expired, you must register the instance and add a valid license to it.
2. If the 90-day evaluation period of a vCenter instance has expired, you must assign a valid license to it from VCF Operations.
3. If the 90-day evaluation period of an ESX host has expired, you must add the host to a vCenter instance that has a valid license with enough available capacity assigned to it.

Note:

After you install VMware Cloud Foundation, it runs in evaluation mode for up to 90 days. If you fail to license your VCF fleet before the evaluation period expires, you must reinstall the software because you cannot license a fleet with an expired evaluation period.

Subscriptions and License Expiration

Multiple subscriptions for the same product can contribute capacity to a single default or split license, and a single subscription can contribute capacity to multiple split licenses.

The total capacity of a default or a split license equals the sum of the license capacity units from all contributing subscriptions that are active. When a subscription expires, its contribution to each license is removed. If other subscriptions contribute capacity to the license, the license remains active but with reduced capacity. When the last contributing subscription expires, the license capacity reaches zero, and the license expires.

VCF subscriptions with different expiration dates.

Name	License Capacity	Start Date	Expiration Date
A	100 cores	July 1, 2027	June 30, 2028
B	25 cores	July 1, 2027	December 31, 2028

You use the license capacity from the two subscriptions to create two split licenses.

- License 1. 75 cores sourced entirely from subscription A. This license expires on June 30, 2028, when Subscription A expires.
- License 2. 50 cores where 25 cores are sourced from subscription A, and 25 cores are sourced from subscription B. Capacity changes over time based on the subscription terms:
  - From July 1, 2027 through June 30, 2028: 50 cores (25 cores from subscription A and 25 cores from subscription B).
  - From July 1, 2028 through December 31, 2028: 25 cores (from subscription B only, subscription A expires).

  License 2 expires on December 31, 2028, when subscription B expires.

For more information about the impact of license expiration on your environment, see Licensing Overview.

High-Level Licensing Workflow

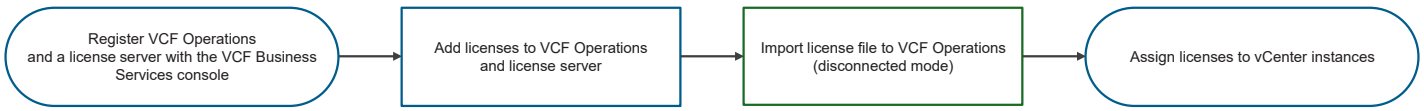
Before you license your environment, you must install a VCF Operations instance of version 9.1 and a license server.

Note:

VCF Operations works in evaluation mode for up to 90 days after you upgrade it from Aria Operations 8.x or after a new deployment of version 9.1. You can license your environment during this period.

If you want to use more than one license server, install the license server instances. For more information, see Deploy a License Server Version 9.1.0.

Figure 394: High-level licensing workflow diagram



1. Register your VCF Operations instance and the license servers with the VCF Business Services console by using one of the two modes:
   - Disconnected mode - no Internet connection is required. You must manually exchange registration, verification, and license files between the VCF Operations instance and the VCF Business Services console.

- Connected mode - Internet connection is required. You use an activation code from the VCF Business Services console to register your VCF Operations instance and the license servers.

  Note:

  Downloading a license file from the Business Services console to VCF Operations is the mechanism for communicating your license configuration to your VCF fleet or vSphere Foundation environment from the Business Services console. This file serves a similar purpose to license keys in pre-version 9.0 releases of VCF. In connected mode, the license file download is automated.

During the registration process, you must add licenses to the license servers from the VCF Business Services console. For more information, see Registering VCF Operations and a License Server with the VCF Business Services Console.

2. Assign licenses to vCenter instances or other assets in your environment from VCF Operations. For more information, see Assign a Primary License to a vCenter Instance, Assign an Add-on License to a vCenter Instance, and Assign an Override License to an ESX Host in VCF Operations.

   Important:

   After you add licenses to the license servers in VCF Operations, you must update the licenses at least once every 6 months (180 days). If license usage data is not submitted within the required reporting time-frame, and you do not update your licenses, your licenses are treated as expired, your hosts are disconnected from the vCenter instance, and you cannot start any workload operations. Existing workloads are not proactively stopped. For more information, see Updating Licenses and Viewing the License Usage File.

   If you want to upgrade a licensed ESX host of version 9.0 to version 9.1, you must first deploy and register a license server.

Mixed-Version Environments

If you have pre-version 9.0 product components, for example, vCenter instances, ESX hosts, NSX, you can continue to use license keys to license capabilities of those components without any changes. If you use a VCF Operations instance for license management, you must update the instance to manage version 9+ licenses.

You can add ESXi 8.x hosts that are licensed with license keys to your vCenter instances of version 9.1. Those ESXi hosts must be licensed with a pre-version 9 license key, and they also use license capacity from your version 9+ license from your VCF or vSphere Foundation subscription. The used license capacity displayed in VCF Operations for this vCenter instance of version 9.1 includes the license capacity used by the licensed ESX hosts of versions 9.0, 9.1, and 8.x that are added to this vCenter instance. Used capacity for the license displayed in the VCF Operations instance and the VCF Business Services console also reflects this. Version 8.x licenses cannot be assigned as override licenses. If you want to upgrade an ESX host of version 9.0 to version 9.1, the host should either have remaining evaluation period, or it must be added to a vCenter instance of version 9.1 that is connected to a license server.

Note:

In a VMware Cloud Foundation Instance, the management domain of version 9.1 can manage workload domains of version 5.x, 9.0, and 9.1.

License usage data is collected for version 9+ licenses, and for licenses of earlier versions connected to VCF Operations. You can view both your version 9+ and pre-version 9 license usage data by using the usage analytics available in VCF Operations and the VCF Business Services console.

When you upgrade your vCenter instances to version 9.1, they operate in evaluation mode for up to 90 days. During that period, you must license the vCenter instances with a primary license with sufficient capacity. The license must be for the product you want to use.

Add-on Licenses Included in Your VCF Subscription

To be able to use an add-on license, you must first assign a primary license to your vCenter instances. Add-on licenses are available for VCF and vSphere Foundation products.

When you purchase VCF, in addition to your VMware Cloud Foundation (cores) license you receive a vSAN tebibytes (TiBs) license. Starting November 3, 2025, based on your VCF subscription, you might also receive a license for VMware Private AI Foundation with NVIDIA.

For information about available add-ons for each product that are not included in your subscription, contact your sales team. Add-ons do not necessarily follow the same licensing model as VCF and vSphere Foundation. You can find information about how to license a specific add-on within their respective documentation.

vSAN Licensing Overview

When you purchase VCF, in addition to your VMware Cloud Foundation (cores) license you receive an equivalent vSAN license for tebibytes (TiBs) of total raw physical storage. You can also purchase additional storage as a separate subscription.

- You can assign a vSAN TiB license to a single vCenter instance or multiple vCenter instances.
- vSAN license usage is based on the storage capacity in the vSAN clusters. vSAN pools the capacity of the subscriptions that you purchased and shares the allocated capacity between multiple vSAN clusters using the vSAN TiB license.
- You do not need a separate license or subscription for vSAN witness appliances. You can deploy as many vSAN witness appliances as required. vSAN witness appliances cannot be used to run workloads not related to vSAN witness.
- Starting with VCF 9.1, you can assign a vSAN TiB license directly to a cluster and override the vCenter license that is automatically assigned to the cluster. For more information about override licenses, see What is an Override License.

vSAN TiB License

When you assign a vSAN TiB license to vCenter, the vSAN license is automatically applied to all the vSAN clusters that are managed by the vCenter. To calculate the capacity that you need for your vSAN environment, you must determine the quantity of TiB licenses required for the total physical device capacity in TiBs on all the ESX hosts in each vSAN cluster.

The vSAN clusters reflect the total raw storage capacity available. For more information about calculating the license capacity that you need for your environment, see Broadcom knowledge base article 95927.

If you have a bare metal host as a vSAN witness, perform the following:

1. Log in to the witness host.
2. Run the following command:
   ```
   esxcli vsan witness license set --enable true
   ```
3. Reboot the witness host.

vSAN License Capacity in VMware Cloud Foundation

If you have a vSAN cluster with 3 ESX hosts, 1 CPU per host, and each host has 6 physical cores per CPU, you must purchase the subscription capacity of 16 cores per CPU as it is the required minimum license capacity. For each VCF cores that you purchase, you receive 1 TiB of vSAN capacity. With a total of 48 (3 * 1 * 16) physical cores per VCF cluster, you receive 48 TiB vSAN capacity. Similarly, with a total of 144 (3 * 2 * 24) physical cores per VCF cluster, you receive 144 TiB vSAN capacity.

Number of ESX hosts (in vSAN cluster)	CPUs per ESX Host	Cores per CPU	Number of Core License	Entitled vSAN Capacity (TIB)
3	1	6 (minimum 16 cores required)	48	48
3	2	24	144	144

To expand the capacity beyond the total entitled tebibytes, you can purchase vSAN add-on capacity. This automatically adds the capacity to the existing default vSAN license.

The license use of vSAN is recalculated and updated in the following cases:

- If you assign a different license to the vSAN cluster

- If you add a new host to the vSAN cluster
- If a host is removed from the vSAN cluster
- If the total number of TiBs in a vSAN cluster changes

vSAN License Capacity in VMware vSphere Foundation

VMware vSphere Foundation license includes an entitlement of 0.25 tebibyte (TiB) of vSAN raw capacity per licensed core. To calculate the capacity that you need for your vSAN environment, you need to determine the total number of licensed CPU cores for each CPU on all the ESX hosts in your environment.

For example, consider a vSAN cluster with 3 ESX hosts, 1 CPU per host, and each host has 8 physical cores per CPU. You must purchase a vSphere Foundation license with the subscription capacity of 16 cores per CPU because it is the required minimum license capacity. With a total of 48 licensed cores (3 hosts * 1 CPU * 16 minimum licensed cores per CPU) , you receive 12 TiBs of included capacity (0.25 TiB *48 licensed cores). Similarly, With a total of 128 licensed cores (4 hosts * 2 CPUs * 16 minimum licensed cores per CPU), you receive 32 TiBs of capacity.

Number of ESX hosts (in vSAN cluster)	CPUs per ESX Host	Cores per CPU	Number of Core License	Entitled vSAN Capacity (TiB)
3	1	8 (minimum 16 cores required)	48	12
4	2	16	128	32

The 0.25 TiB of vSAN capacity per licensed core pools with any vSAN capacity from other add-ons or VCF purchases. You can use this capacity to license the vSAN clusters in their vCenter instances, regardless of whether the capacity is more or less than 0.25 TiB per core.

License Server Overview

A license server is added to a VCF Operations instance to store the licenses you want to assign from this VCF Operations instance. You must add at least one license server to each VCF Operations instance that you use for license management.

Licenses for VCF infrastructure and management components are stored on a dedicated license server appliance. The license server is a protected repository within your environment to keep your licensing data.

A license server is deployed during new VCF or vSphere Foundation deployments, and during upgrades to version 9.1 and later.

You must add at least one license server to each VCF Operations instance that you use for license management. Most VCF Operations instances only need one license server. In some cases you might need additional license servers. If you have a vCenter instance that does not have reliable connectivity to your data center, the connectivity latency exceeds 300 ms, you can deploy a license server in that vCenter instance to avoid licensing-related service interruptions. For example, if you have an edge site with limited connectivity to your management domain, you might not be able to license new hosts added to your edge site.

Each license server must have a unique license allocation. There must be a connection between the license server and VCF Operations to complete a license refresh at least once every 180 days. To assign a license to an ESX host, the vCenter instance that the host is added to must be able to connect to the license server.

You can use a single license server to license your entire fleet, but you cannot share a license server between fleets.

After you add a license server to VCF Operations, you must add licenses to the license server to be able to license your vCenter systems and other assets in your environment.

License Server Models

- Single license server model. This model provides a single, unified repository for the entire fleet. It is suitable for deployments where reduced operational overhead is prioritized.

- Multi license server model: This model supports environments that require administrative division for multi-tenancy or geographic distribution. It ensures local license availability for regional sites that lack reliable connectivity to the central management domain.

Sizing Requirements and Scale

Sizing requirements:

- 2 vCPUs
- 4-GB RAM
- 8-GB Disk

Scale: 250,000 assets

All licensed vCenter instances, hosts, and vSAN clusters are assets.

Monitoring License Server Health

You can monitor the health of a license server in VCF Operations, on the details page of the specific license server.

For detailed monitoring information, you can view the License Server Health dashboards in VCF Operations. For more information about using VCF Operations health monitoring dashboards, see Using Predefined Dashboards for Quick Insights.

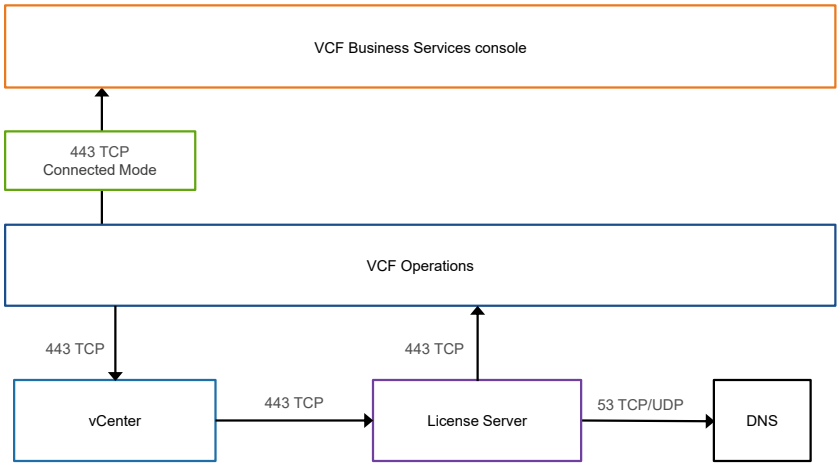
License Server Networking

The license server operates within a strictly defined network perimeter and does not communicate with the Internet. There are no external network requirements for the license server. To ensure reliable operations, the network path must support a maximum latency of 300 ms.

Ports and Protocols Used for Licensing-Related Operations

Figure 395: Licensing Connectivity Diagram

The diagram illustrates the license server networking connectivity with other VCF components.



Source	Destination	Description
vCenter	License server	When a license is assigned to a vCenter instance, or when a host is added to or removed from a licensed vCenter instance, the vCenter instance connects to the license server.
License server	VCF Operations	The license server regularly connects to VCF Operations to check for updates, provide license usage data, and retrieve updated license files.
License server	DNS servers	The license server connects to the specified DNS servers for name resolution.
VCF Operations	vCenter	VCF Operations connects to vCenter instances to coordinate the assignment of available licenses to the vCenter instances.

No other components require a network connection to the license server. For a complete list of network ports and protocols required for licensing-related operations, go to VMware Ports and Protocols and enter *license* in the *Search Listings* search box.

If your VCF Operations instance is registered in disconnected mode, you must manually submit aggregated license usage data. If your VCF Operations instance is registered in connected mode, it periodically communicates aggregated license usage data to Broadcom

Network Communication Between VCF Operations and Broadcom

In connected mode, VCF Operations periodically communicates aggregated license usage data to Broadcom. For a complete list of URLs and network ports, see Public URLs Required for Online Functionalities for VCF and vSphere Foundation.

Internal Network Requirements

For versions 9.1.0.0 and 9.1.0.0100, the license server uses an internal communications network in the 172.16.0.0/12 and 172.17.0.0/16 ranges. To avoid conflicts and allow for normal work of the license server, do not deploy the license server on the 172.16.0.0/12 network. Also do not configure the license server to use a gateway that advertises a routable path of 172.17.0.0/16.

Starting with version 9.1.0.0200, the license server works with a broader range of IP addresses for the internal communication. The limitations that are listed above no longer apply.

License Server IPv6 Support

Starting with version 9.1.1, the license server supports IPv6 networking that uses static addressing, or DHCPv6/SLAAC. The method for configuring IPv6 is different depending on your scenario.

1. New deployment by using VCF Installer. VCF Installer only deploys the license server with IPv4. To configure IPv6 on the license server, use the vSphere Client. For more information, see Configure IPv6 on a License Server.
2. New deployment by using an OVA file. During the deployment, you can configure the license server for IPv4-only, IPv6-only, or dual-stack networking. For more information, see Deploy a License Server Version 9.1.1.
3. License servers patched to version 9.1.1. You cannot activate IPv6 on the license server. To use IPv6, you must deploy a new license server of version 9.1.1 and transfer your licenses to it. For more information, see Deploy a License Server Version 9.1.1, Register an Additional License Server with the VCF Business Services Console, and Transfer Licenses After You Upgrade VCF Operations from Version 9.0.x.

Configure IPv6 on a License Server

Starting with version 9.1.1, the license server supports IPv6 networking. You can select a dynamic address allocation DHCPv6/SLAAC, or you can manually assign a static IPv6 address, prefix length, and default gateway.

- Required privileges: **VApp.ApplicationConfig** and **VApp.InstanceConfig** on the license server virtual machine.
- Power off the license server virtual machine.

> **Note:**
>
> If you update a license server from version 9.1.0 to version 9.1.1, you cannot configure IPv6 on that license server instance. To utilize IPv6 networking, you must deploy a new license server instance of version 9.1.1 and transfer your licenses to it. For more information, see Deploy a License Server Version 9.1.1, Register an Additional License Server with the VCF Business Services Console, and Transfer Licenses After You Upgrade VCF Operations from Version 9.0.x.

The license server supports two IPv6 configuration modes:

- Static. You manually configure the default gateway, prefix length, and IP address.
- DHCP/SLAAC. The default gateway, prefix length, and IP address are configured automatically.

1. Log in to the vSphere Client, and navigate to the license server virtual machine.
2. Select the virtual machine, and click the **Configure** tab.
3. Under **Settings**, select **vApp Options**.
4. Navigate to the **Properties** section.
5. Select the IPv6 assignment method, and configure the required property values.
   - For dynamic addressing, select DHCP/Slaac. The license server automatically obtains its IPv6 address, prefix length, and default gateway.
   - For static addressing, select static. You must manually configure the static network details.

Option	Description
Configure dynamic addressing (DHCPv6/SLAAC)	1. In the **Key** column, select **ipv6_type**, and click **Set Value**. 2. From the Property Value drop-down menu, select **DHCP/ Slaac**, and click **OK**.
Configure static addressing	1. In the **Key** column, select **ipv6_type**, and click **Set Value**. 2. From the Property Value drop-down menu, select **Static**, and click **OK**. 3. In the **Key** column, select **ipv6_address.VCF_License_Server_Appliance**, and click **Set Value**.

Option	Description
	4. Configure the IP address, and click **OK**. 5. In the **Key** column, select **ipv6_gateway.VCF_License_Server_Appliance**, and click **Set Value**. 6. Configure the Default Gateway, and click **OK**. 7. In the **Key** column, select **ipv6_netmask.VCF_License_Server_Appliance**, and click **Set Value**. 8. Configure the Prefix Length, and click **OK**.

6. Configure DNS for IPv6.
   a) In the **Key** column, select **dns.VCF_License_Server_Appliance**, and click **Set Value**.
   b) Enter the IPv6 addresses of your DNS servers, and click **OK**.
7. Set the network stack mode.

   Note:

   By default, after you configure IPv6, the license server operates in dual-stack mode. If you want to disable IPv4 and restrict the appliance to IPv6-only, configure the **forceIpv6** value.

   a) In the **Key** column, select **forceIpv6**, and click **Set Value**.
   b) Select the check box For IPv6 only, and click **OK**.
8. To apply the IPv6 configuration, power on the license server virtual machine.

The license server is configured to use IPv6.

License Server Security

A license server is an inherently sensitive component of any VMware Cloud Foundation deployment. The license server is hardened and configured to the strictest STIG requirements.

The license server retains a record of licenses and assets to which a license is assigned. The license server retains the metadata and license data applicable to those licenses and assets and does not retain other data. The license server does not scan your environment.

The license server communicates through the port and protocols that are listed at VMware Ports and Protocols. All other inbound and outbound traffic can be blocked.

VCF Operations handles all other licensing-related communication. For example, in connected mode, VCF Operations periodically communicates aggregated license usage data to Broadcom. In disconnected mode, you must manually submit that data. For more information, see Updating Licenses and Viewing the License Usage File.

The license server uses the time set on the host which contains the license server virtual machine. To avoid license assignment failure, ensure that the time on the ESX host which contains the license server virtual machine is synchronized with the time set on the other VCF components.

The license server does not support SSH connections, and you cannot enable SSH on the license server. Root or other user accounts are not present on the license server.

License Server Downtime Impact

If the license server is unavailable, you cannot assign licenses, generate license usage reports, or import license files.

The license server does not need to be always online. There is no immediate workload or management operations impact if the license server becomes unreachable by VCF Operations or vCenter instances. To minimize potential downtime, ensure that the cluster which contains the license server virtual machine is configured with vSphere High Availability.

While the license server is unavailable to VCF Operations, the following operations are impacted:

- Assigning licenses to new vCenter instances: You can add new vCenter instances in evaluation mode. They are fully operational during the evaluation period.

- Generating license usage reports: Updated license usage reports cannot be generated until connectivity to the license server is restored. If VCF Operations is registered in connected mode, the daily automated license usage reports cannot be generated. After the 180-day update period, the licenses expire.
- Importing license files: You cannot add licenses to the inventory. Prior license usage reports cannot be confirmed and the license update period cannot be refreshed.

While the license server is unavailable to vCenter , when you add new hosts to the vCenter instance, the hosts cannot be licensed. You can add new ESX hosts in evaluation mode. They are fully operational during the evaluation period.

While the license server is unavailable to vCenter , when you add new hosts and enable vSAN clusters in the vCenter instance, the hosts and vSAN clusters cannot be licensed. You can add new ESX hosts and vSAN clusters in evaluation mode. They are fully operational during the evaluation period.

License Server Recovery

A license server outage does not impact running workloads or management operations until your license usage report due date expires or a new license assignment is required. Because license assignment data is restored by VCF Operations, you do not need to perform manual backups or repairs on license servers. If a failure occurs, you can deploy, register, and transfer your licenses to a new license server. VCF Operations restores all previous license assignment data to the new license server instance.

Due to integrated security controls, you cannot restore a license server from a backup or a snapshot, and you cannot clone the license server. To restore the license server and its data:

1. Deploy a new license server appliance and register it with the VCF Business Services console. For more information, see Deploy a License Server Version 9.1.0.
2. In the VCF Business Services console, delete the unavailable license server registration. For more information, see Delete a License Server.
3. In the VCF Business Services console, add the licenses that were added to the deleted license server to the newly deployed license server.

> **Note:**
>
> After you import the license file in VCF Operations, the licenses are automatically reassigned to the same assets that they were assigned to before you deleted the license server.

Registering VCF Operations and a License Server with the VCF Business Services Console

You access and manage the licenses for your environment by using VCF Operations, and the VCF Business Services console. You must register the VCF Operations instance and a license server with the VCF Business Services console.

You must install a VCF Operations instance version 9.1 to manage licenses for VCF components of version 9.1. If you want to access and manage your 9.1 licenses, and you already have an earlier version of VCF Operations installed, you must upgrade the instance to version 9.1.

A license server is automatically deployed as part of the installation and upgrade of VCF. Each license server contains licenses for a specific VCF Operations instance. You can add licenses for different products in the same license server, and you can add as many licenses to a license server as you need. You can add licenses to the license server at any time after the initial registration process. For more information, see Add a License to VCF Operations and License Server.

> **Note:**
>
> After the deployment of the license server as part of an upgrade from 9.0.x to 9.1, your licenses are transferred from the VCF Operations instance to the license server.
>
> If your VCF Operationsinstance is in connected mode, the license server is automatically registered and licenses are automatically transferred to the license server.

If your VCF Operationsinstance is in disconnected mode, you must register the license server with the VCF Business Services console. Licenses are then automatically transferred from VCF Operations to the license server. For more information, see Transfer Licenses After You Upgrade VCF Operations from Version 9.0.x.

You need at least one license server and a VCF Operations instance to license your environment. If you need another license server, you can download and install a license server manually. For more information, see Deploy a License Server Version 9.1.0.

After you deploy or upgrade your VCF Operations instance to version 9.1, it works in evaluation mode for up to 90 days. During that period you must register VCF Operations and the license server in the VCF Business Services console, and you must license your environment.

Note:

If you want to use a VCF Operations instance only for monitoring your assets, and not for license management, you do not need to register it with the VCF Business Services console. A VCF Operations instance that you use only for monitoring purposes is automatically licensed with the primary license assigned to the vCenter instances of version 9.0 or 9.1 that are added to the VCF Operations instance. These vCenter instances must be licensed by a different VCF Operations instance.

You can use specific VCF Operations capabilities based on the primary license that is assigned to each vCenter.

Product	VCF Operations Capabilities
VMware Cloud Foundation (includes VCF Edge)	License Management, Monitoring, Fleet Management
vSphere Foundation	License management, Monitoring

One VCF Operations instance can manage a mix of different license types and licenses for different products. For example, you can manage both VCF and VMware Foundation primary licenses in the same VCF Operations instance, and add-on licenses for different product components.

You can choose whether to register your VCF Operations instance in connected or disconnected mode.

- In disconnected mode, to register VCF Operations, you must manually transfer files between the VCF Operations instance and the VCF Business Services console. After the registration, you must manually transfer files to report license usage, and to update licenses.
- In connected mode, VCF Operations registration does not require manual transfer of files, the registration is performed by using an activation code. After the registration, usage reporting is automated, and the update of licenses is simplified.

CAUTION:

To use VCF Operations for licensing, verify that the Virtual Management Infrastructure (VIM) adapter is running.

By default the fully qualified domain name (FQDN) of VCF Operations is prepopulated as a display name when you start the registration process in the VCF Business Services console. The FQDN is only saved if you do not enter another display name. If you change the FQDN to another display name, the connectivity between VCF Operations and the VCF Business Services console is not impacted. You can change the display name at any time after the registration.

If you do not want to include the FQDN of the VCF Operations instance or the hostname of the license server in the registration details, you can use the following API before you start the registration process:

```
PUT {VCF_Ops_URI}/suite-api/api/deployment/config/globalsettings/GENERATE_DEFAULT_LICENSE_MANAGER_ASSET_NAME/
false
```

How to view the detailed contents of your registration file

If your VCF Operations instance is in connected mode, the registration details are sent automatically by using a combination of query string parameters and API parameters. If your VCF Operations instance is in disconnected mode, you must manually download the registration information as a JSON Web Signed (JWS) file. The information provided to Broadcom for registration is equivalent between connected and disconnected mode, but sent in different formats.

Note:

You can view the registration file contents only in disconnected mode.

1. Before you complete the registration process, save a copy of your registration file.
2. Use a text editor to open the file, and copy the content.
3. Paste the content of the registration file into a JWT decoding tool.

After the content is decoded, you can view the collected registration information.

Table 980: Registration File Content

Property	Applies to version	Description
model_version	9.0 and later	File format version: Always *1.0.0*.
product_version	9.1 and later	The version of the VCF Operations instance.
asset_name	9.0 and later	The FQDN of the VCF Operations instance. The FQDN is pre-populated as a display name for the instance in the VCF Business Services console. The FQDN is only saved by Broadcom if you choose to keep the display name the same as the FQDN.
created_on	9.0 and later	Date and time when the registration file was created.
asset_type	9.0 and later	Always *AC*.
xr2	9.0 and later	A unique opaque fingerprint of the VCF Operationsinstance. It is used in conjunction with the asset_id to link subsequent usage data to the registered VCF Operations instance. No identifiable details about your environment can be derived from the fingerprint.
asset_id	9.0 and later	The unique ID of the VCF Operations instance you register with this file.
request_id	9.0 and later	Unique ID of the registration file.
child_assets	9.1 and later	The list of the license servers associated with the VCF Operations instance.
child_assets.asset_id	9.1 and later	The unique ID of this license server instance being registered.
child_assets.asset_name	9.1 and later	The hostname of the license server instance. The hostname is pre-populated as a display name for the instance in the VCF Business Services console. The hostname is only saved by Broadcom if you choose to keep the display name the same as the hostname.
child_assets.asset_type	9.1 and later	Always *LS*.
child_assets.product_version	9.1 and later	The version of this license server instance.

Starting with version 9.1, to complete the registration process, you must also provide a confirmation file. It lists all license servers in the environment.

Table 981: Confirmation File Content

Property	Applies to version	Description
child_assets	9.1 and later	The list of the license servers associated with the VCF Operations instance.
child_assets.asset_id	9.1 and later	The unique ID of this license server instance being registered.
child_assets.asset_type	9.1 and later	Always *LS*.
child_assets.product_version	9.1 and later	The version of this license server instance.
child_assets.public_key	9.1 and later	The public key of the license server.
child_assets.fingerprint	9.1 and later	A unique opaque fingerprint of the license server instance being registered. It is used in conjunction with the asset_id to link subsequent usage data to the registered license server instance. No identifiable environment details can be derived from the fingerprint.

Property	Applies to version	Description
child_assets.challenge	9.1 and later	A unique code used as part of the confirmation process to validate the license server.
child_assets.timestamp	9.1 and later	The date/time when the confirmation file was created.
child_assets.expiry	9.1 and later	The date and time when the confirmation file expires. The file is valid for 3 hours after it is generated. You must submit the confirmation file before the specified expiration date and time.
asset_id	9.1 and later	The version of the VCF Operations instance.
request_id	9.1 and later	A unique ID of the confirmation file.

Register VCF Operations and Your License Server in Connected Mode

You must register the VCF Operations instance that you want to use for license management and at least one license server with the VCF Business Services console. If your environment is connected to the Internet, you can register your VCF Operations instance in connected mode. Connected mode provides a faster and simplified registration process, and makes updating your licenses easier. Connected mode is recommended for anyone who has an Internet connection.

- Verify that you have a valid subscription for VCF or vSphere Foundation.
- Verify that your VCF Operations instance is connected to the Internet. If you must configure an Internet proxy, configure the proxy in VCF Operations, under **Global Settings** > **Network Settings**.
- Verify that you have a Broadcom Support Portal role or a VCF Business Services console role with privileges to register the VCF Operations instance. For more information about the role mappings, see Access Management.

In connected mode, the update of licenses can be fully automated or semi-automated. For more information, see Change License Updates Mode.

Note:

By default the FQDN of VCF Operations is prepopulated as a display name when you start the registration process in the VCF Business Services console. The FQDN is only saved if you do not enter another display name. If you change the FQDN with another display name, the connectivity between VCF Operations and the VCF Business Services console is not impacted. You can change the display name at any time after the registration.

If you do not want to include the FQDN in the registration details, you can use the following VCF Operations API before you start the registration process:

```
PUT {VCF_Ops_URI}/suite-api/api/deployment/config/globalsettings/
GENERATE_DEFAULT_LICENSE_MANAGER_ASSET_NAME/false
```

A license server is automatically deployed during a new VCF Operations 9.1 deployment and an upgrade to VCF Operations 9.1.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, click **Licensing** > **Licenses & Registration**.
3. In the **Register & License VCF Operations** pane, click **Continue**.

   The **Register and License VCF Operations** workflow appears.

   The first step of the procedure is complete because a license server is automatically deployed during a new VCF Operations 9.1 deployment and an upgrade to VCF Operations 9.1.

4. In the **Select Connection Mode** card, select **Start**, and click **Continue**.
5. In the **Select Connection Mode** page, select **Connected**, and click **Continue**.
6. Navigate to the **Registration** section.
7. Get an activation code for your registration.
   a) In the **Get Activation Code from VCF Business Services console** card, click **Start**.

      The VCF Business Services console opens in a new tab.
   b) Log in to the VCF Business Services console by using your Broadcom Support Portal credentials.
   c) Select the Site ID to which you want to register this VCF Operations instance, and click **Next**.

      If you have only one site, it is selected by default.
   d) In the VCF Business Services console, on the **Register & License VCF Operations** page, navigate to the **Registration** part of the registration workflow.
   e) In the **Name VCF Operations** pane, click **Start**, enter a unique display name for your VCF Operations instance, and click **Save**.

      The FQDN is only saved if you do not enter another display name. If you change the FQDN with another display name, the connectivity between VCF Operations and the VCF Business Services console is not impacted. You can change the display name at any time after the registration.
   f) In the **Generate Activation Code** card, click **Start**.

      A dialog box with the activation code appears.
   g) Click **Copy**.
   h) Verify that you copied the code, and click **Finish**.
8. Navigate to the VCF Operations instance, and upload the activation code.
   a) In VCF Operations, in the **Paste Activation Code** card, click **Start**.

      A dialog box appears.
   b) In the text field, paste the activation code, and click **Activate**.
9. In the **Add Licenses from VCF Business Services console** card, click **Start**.

   The VCF Business Services console opens in a new tab.
10. In the VCF Business Services console, add licenses to your license servers.

    Note:

    To complete the registration process, you must first add licenses to each license server.

    a) Select the Site ID to which you want to register this VCF Operations instance, and click **Next**.

       The site that you selected in the beginning of the registration flow is selected by default.
    b) In the Add Licenses section of the registration workflow, in the card for the respective license server, click **Start**.
    c) On the **Allocate Licenses** page, enter a display name for the license server.
    d) If you want to split the default license or change the capacity of a license before you assign it, click the vertical ellipsis button next to the license.
       1. To split the default license, select it from the list and click **Create**.
          a. Enter a name and license capacity for the split license.
       2. To change the capacity of a split license, enter the capacity for the license in the New Capacity column.
       3. Click **Save**.
    e) Select licenses from the table to add to the license server, and click **Next**.

       Note:

       To license your environment, you must add at least one primary license to the license servers. You can add licenses during the registration process, or at any time after that.

11. Download the license file in the VCF Operations instance.
    a) Navigate to the **Licenses** page.
    b) In the **Add Licenses to License Servers** section, in the **Download** card, click **Start**.

       A dialog box appears.
    c) To download the license file, click **Save**.
12. On the **Register and License VCF Operations** page, click **Finish**.

- To license your environment, assign licenses to your vCenter instances. For information on how to assign a license, see Assign a Primary License to a vCenter Instance.
- Update the licenses at least once every 180 days. For more information about updating licenses, see Update Licenses in Connected Mode. If you want to switch to automated update of licenses, see Change License Updates Mode.

Register VCF Operations and Your License Server in Disconnected Mode

You must register the VCF Operations instance that you want to use for license management with the VCF Business Services console. If your environment is disconnected from the Internet, you can register your VCF Operations instance in disconnected mode.

- Verify that you have a VCF Operations instance version 9.1.
- Verify that you have a valid subscription for a VMware product.
- Verify that you have a Broadcom Support Portal role or a VCF Business Services console role with privileges to register the VCF Operations instance. For more information about the role mappings, see Access Management.

After you register your VCF Operations instance in disconnected mode, you must manually submit usage reports in the VCF Business Services console. Then you must update your licenses by importing a license file in the VCF Operations instance.

Note:

By default the FQDN of VCF Operations is prepopulated as a display name when you start the registration process in the VCF Business Services console. The FQDN is only saved if you do not enter another display name. If you change the FQDN with another display name, the connectivity between VCF Operations and the VCF Business Services console is not impacted. You can change the display name at any time after the registration.

If you do not want to include the FQDN in the registration details, you can use the following VCF Operations API before you start the registration process:

```
PUT {VCF_Ops_URI}/suite-api/api/deployment/config/globalsettings/
GENERATE_DEFAULT_LICENSE_MANAGER_ASSET_NAME/false
```

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, click **Licensing** > **Licenses & Registration**.
3. In the **Register & License VCF Operations** pane, click **Continue**.

   The **Register and License VCF Operations** workflow appears.

   The first step of the procedure is complete because a license server is automatically deployed during a new VCF Operations 9.1 deployment and an upgrade to VCF Operations 9.1.

4. In the **Select Connection Mode** card, select **Start**, and click **Continue**.
5. On the **Select Connection Mode** page, select **Disconnected**, and click **Continue**.
6. Download the registration file.
   a) In the **Registration** section, in the **Download Registration File** card, click **Download**.
   b) Save the registration file.
7. Use a computer with Internet connection and log in to the VCF Business Services console with your Broadcom Support Portal credentials.
   a) In the VCF Operations instance, in the **Import Registration File to the VCF Business Services console** card, click **Start**.

      A new tab opens in your browser, and you are redirected to the VCF Business Services console at https://vcf.broadcom.com/.
   b) Enter your Broadcom Support Portal credentials.
   c) Select the Site ID to which you want to register this VCF Operations instance.

      If you have only one site, it is selected by default.
8. Import the registration file in the VCF Business Services console.
   a) Navigate to **Licensing** > **VCF Operations Registrations**.
   b) In the **License Your Environment** card, click **Upload Registration File**.
   c) On the VCF Operations Registrations page, click **New Registration**.

      The **Register & License VCF Operations** workflow appears.
   d) In the **Registration** section, in the **Import Registration File** card, click **Start**.

      A dialog box appears.
   e) In the **Import Registration File** dialog box, click **Browse**, and select and import the license file.

      Note:

      If you upload a registration file that is already used to register another VCF Operations instance, the operation fails with an error.
9. In the VCF Business Services console, enter a display name for your VCF Operations instance.
   a) In the Registration section, in the **Name VCF Operations** card, click **Start**.
   b) Enter a unique name for your VCF Operations instance, and click **Save**.

      By default the FQDN of VCF Operations is prepopulated as a display name when you start the registration process in the VCF Business Services console. The FQDN is only saved if you do not enter another display name. If you change the FQDN with another display name, the connectivity between VCF Operations and the VCF Business Services console is not impacted. You can change the display name at any time after the registration.
10. Complete the verification process by exchanging the verification and confirmation files between the VCF Business Services console and the VCF Operations instance.
    a) In the Registration section, in the **Download Verification File** card, click **Start**, and save the verification file.
    b) In your VCF Operations instance, in the **Import Verification File** card, click **Import**, and upload the verification file.

       After you import the verification file, the confirmation file is ready for download.

c) In the **Download Confirmation File** card, click **Download** to download the confirmation file.
d) To go back to the VCF Business services console, in the **Upload the Confirmation File to the VCF Business Services console** card, click **Start**.
e) To verify that you uploaded the verification file in VCF Operations, in the Upload the Verification file to VCF Operations card, click **Mark as Complete**.
f) In the **Import Confirmation File** card, click **Start**, and upload the confirmation file that you downloaded from VCF Operations.

11. In the VCF Business Services console, add licenses to your license servers.

    Note:

    To complete the registration process, you must first add licenses to each license server.

    a) In the Add Licenses section of the registration workflow, in the card for the respective license server, click **Start**.
    b) In the **Add Licenses** wizard, enter a display name for the license server.
    c) If you want to split the default license or change the capacity of a license before you assign it, click the vertical ellipsis button next to the license.
        1. To split the default license, select it from the list, and click **Create**.
            a. Enter a name and license capacity for the split license.
        2. To change the capacity of a split license, enter the capacity for the license in the New Capacity column.
        3. Click **Save**.
    d) Select licenses from the table to add to the license server, and click **Next**.

    Note:

    To license your environment, you must add at least one primary license to the license servers. You can add licenses during the registration process, or at any time after that.

12. In the Download license file card, click **Start**, and download the license file.

    You must import the license file in the VCF Operations instance to complete the registration process.

13. In the VCF Operations instance, import the license file that you downloaded from the VCF Business Services console.
    a) In the **Import License File** card, click **Import**, and import the licenses file that you downloaded from the VCF Business Services console.

    A dialog box appears.
    b) In the **License File Imported** dialog box, review the licenses imported to the VCF Operations instance, and click **Close**.

14. To close the Registration page in VCF Operations, click **Finish**.

15. Navigate to the VCF Business Services console, and in the **Import License File to VCF Operations** card, click **Mark as Complete**.

16. At the bottom of the **Register & License VCF Operations** page, click **Close**.

- To license your environment, assign licenses to your vCenter instances. For information on how to assign a license, see Assign a Primary License to a vCenter Instance.
- Update the licenses at least once every 180 days. For more information about updating licenses, see Update Licenses in Connected Mode.

Generate a New Activation Code for VCF Operations Registration with the VCF Business Services Console

If you register VCF Operations with the VCF Business Services console in connected mode, you must use an activation code generated in the VCF Business Services console for the specific registration. If you lose this activation code, you can generate a new one.

- Verify that your VCF Operations instance is connected to the Internet. If you must configure an Internet proxy, see Configuring HTTP Proxy for Outbound Settings.

- Verify that you have a VCF Business Services console role with privileges to register the VCF Operations instance. For more information about the role mappings, see Access Management.

1. Log in to the VCF Business Services console and click VCF Home.

   A list with your incomplete VCF Operations registrations appears.
2. Select the VCF Operations instance for which you want to generate a new activation code.
3. In the **Generate Activation Code** card, click **Start**.

   A dialog box with the activation code appears.
4. Click **Copy**, and click **Finish**.

Register an Additional License Server with the VCF Business Services Console

After you deploy an additional license server and add it to VCF Operations, you must register the license server with the VCF Business Services console.

- Deploy a license server and add it to VCF Operations. For more information, see Deploy a License Server Version 9.1.0.
- Verify that you have a Broadcom Support Portal role or a VCF Business Services console role with sufficient privileges to register a license server. For more information about the role mappings, see Access Management.

If the VCF Operations instance to which you added the license server operates in connected mode, you register the license server in connected mode.

If the VCF Operations instance to which you added the license server operates in disconnected mode, you register the license server in disconnected mode.

You must add licenses to the license server before you can assign them to the assets in your environment.

Register in Connected Mode

1. In the VCF Operations UI, in the red banner on top of the page click **Register License Server**. The **Manage Licenses** page appears.
2. In the card with the newly added licenses server, click **Start**.
3. Optional: On the **Add Licenses** page, enter a display name for the license server.
4. Select licenses from the table to add to the license server and click **Next**.
5. In VCF Operations, update your licenses.
6. After you update your licenses, in the VCF Business Service console, in the **Update Licenses in VCF Operations** card, click **Mark as Complete**.
7. Click **Close**.

Register in Disconnected Mode

1. In the VCF Operations UI, in the red banner on top of the page click **Register License Server**. The **Register License Server** page appears.
2. In the **Download Registration File** card, click **Start**, and save the registration file.
3. To go to the VCF Business Services console and upload the registration file, in the **Upload Registration File** card, click **Start**.
   You are redirected to the **Add License Servers** page in the VCF Business Services console.
4. In the Import Registration File card, click **Start** and import the registration file.
   The Registration section is updated with the name of the VCF Operations instance and the verification cards.
5. In the **Download Verification File** card, click **Start** and save the verification file.
   The verification and confirmation files that are exchanged between the VCF Operations instance and the VCF Business Services console verify the integrity of the license server.
6. In VCF Operations, in the **Import Verification File** card, click **Start**, and import the verification file that you downloaded from the VCF Business Services console.
   A confirmation file is generated.
7. In the **Download Confirmation File** card, click **Start** and save the confirmation file.
8. In the VCF Business Service console, confirm that you uploaded the verification file in the VCF Operations instance and then import the confirmation file.
9. In the **Upload Verification File to VCF Operations** card, click **Mark as Complete**.

10. In the **Import Confirmation File** card, click **Start** and import the confirmation file that you downloaded from VCF Operations.
11. In the Add Licenses section, in the card with the newly added licenses server, click **Start**.
12. Select licenses from the table to add to the license server and click **Next**.
    a. Download the license file from the VCF Business Services console and import it in the VCF Operations instance.
    b. In the VCF Operations instance, in the Import License File card, click **Start**, and import the new license file.
    c. In the **Download License File** card, click **Start** and save the new license file.
    d. To complete the registration, click **Finish**.
    e. In the VCF Business Services console, in the **Import License File to VCF Operations** card, click **Mark as Complete**.
13. After you complete the registration flow, in the VCF Business Services console, click **Close**.

Switch from Connected to Disconnected Mode

You can switch from connected to disconnected mode at any time. In disconnected mode, you must manually submit license usage reports and update your licenses.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, click **Licensing** > **Licenses & Registration**.
3. In the Registration and License Server Status card, click **Manage** > **Registration Details**.

   The **VCF Operations Registration** page appears.
4. On the **Registration Details** tab, next to Reporting Mode, click **Switch to Disconnected.**
5. In the **Switch Reporting Mode** dialog box, click **Switch to Disconnected**.

Your VCF Operations instance operates in disconnected mode. You must manually report license usage, and then upload license files in VCF Operations. For more information about reporting license usage and updating your licenses, see Report License Usage and Update Licenses in Disconnected Mode.

Switch from Disconnected to Connected Mode

You can switch from disconnected to connected mode at any time. In connected mode, license usage reports are submitted automatically, and the update of licenses is easier.

To switch from disconnected to connected mode, verify that your environment is connected to the Internet.

If your VCF Operations instance operates in disconnected mode for more than 180 days, switching from disconnected to connected connection might fail, and you must generate a new activation code. For information on how to generate a new activation code, see Register VCF Operations and Your License Server in Connected Mode.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, click **Licensing** > **Licenses & Registration**.
3. In the Registration and License Server Status card, click **Manage** > **Registration Details**.

   The **VCF Operations Registration** page appears.
4. On the **Registration Details** tab, next to Reporting Mode, click **Switch to Connected.**
5. In the **Switch Reporting Mode** dialog box, click **Switch to Connected**.

Your VCF Operations instance operates in connected mode. License usage is reported automatically. You must update your licenses on regular intervals. For more information about updating your licenses, see Update Licenses in Connected Mode.

Managing Licenses in the VCF Business Services Console

Starting with version 9.0, you have one default license per product. You can split the default license into as many licenses as you need within your total purchased capacity. When you delete a split license, or reduce its capacity, the capacity is added back to the capacity of the default license. You can rename a license at any time.

User Roles Required for License Management

You must have the required permissions to manage licenses in the VCF Business Services console. For more information about the roles in the VCF Business Services console, see Access Management.

Split Default Licenses

You can split default licenses into as many licenses as you need. You can allocate part of the default license capacity to each of the new licenses that you create as a result of the split.

When you create a split license, the capacity you add to it is extracted from the available capacity in the default license. If the available capacity in the default license is less than the capacity that you want to add to the new split license, you can reduce the capacity of other split licenses. The capacity that you deduct from split licenses is added back as available capacity in the default license.

1. In the VCF Business Services console, navigate to **Licensing** > **Licenses**.
2. Click the **Version 9+** tab.
3. Click **Manage Licenses (Split/Merge)**.

   The **Manage Licenses** wizard appears.
4. From the drop-down menu, select a product for which you want to split the default license.

   The default license and all previous split licenses for this product appears.
5. Click **Create**.

   A new line for a split license appears in the table.
6. In the **Name** column, enter a name for the license.
7. In the **New Capacity**column, enter the amount of capacity you want to allocate to the new license.

   You cannot allocate a higher amount of capacity than what is available in the default license.
8. To create more split licenses, repeat steps 5 through 7.
9. Click **Next**.
10. On the **Summary** page, review your changes, and click **Finish**.

Your newly created license appears in the list of licenses in the VCF Business Services console.

If your default license is added to a VCF Operations instance, you must update the licenses in that VCF Operations instance after the split. For more information about updating your licenses, see Updating Licenses and Viewing the License Usage File.

To rename the license, click the vertical ellipsis next to the license, and select **Rename**.

Merging Licenses

When you delete a split license, or reduce its capacity, the capacity you remove from the split license is merged back into the default license.

- For more information about deleting a license, see Delete a License.
- For more information about removing license capacity from a split license, see Change License Capacity.

Change License Capacity

You can change the license capacity of split licenses as many times as you need. You can allocate more capacity or remove some of the capacity of the split licenses at any time. Any change in the capacity allocated to split licenses affects the default license.

The capacity that you add to split licenses is extracted from the capacity of the default license. The capacity you remove from split licenses is added back to the default license. You cannot add more capacity to a split license than the capacity available in the default license. If you need more capacity than what is available in the default license, you must free up capacity from the other split licenses which belong to the same product type, or you must purchase more capacity.

You cannot reduce the capacity of a license below the quantity of the license that is currently used. If you try to reduce the capacity of an assigned license below the used capacity, the operation fails. For example, if you have a license with capacity of 250 cores, and it is assigned to a vCenter instance, which has 4 ESX hosts added to it, each using 50 cores of license capacity, you cannot reduce the capacity of the license to less than 200 cores.

> **Note:**
>
> If at a certain point the used capacity of a license is greater than the capacity allocated to the license, you must allocate more capacity to that license. Until you add more capacity, you cannot assign the over used license to additional vCenter instances. If a vCenter instance is licensed by a license that is over used, you can only add new ESX hosts if they are in evaluation mode.

1. In the VCF Business Services console, navigate to **Licensing** > **Licenses**.
2. Click the vertical ellipsis next to the split license, and select **Change Capacity**.
3. In the **New Capacity**column, enter the new amount of allocated capacity.

   For example, if your license has 150 cores of allocated capacity, and you want to add 50 cores, you add 200 in the New Capacity column.

   You cannot assign more capacity than what is available in the default license.
4. Click **Next**.
5. On the Summary page, review your changes, and click **Finish**.

The new capacity of the modified license and the default license appear in the list of licenses.

Remove a License from a License Server and the Respective VCF Operations Instance

To remove a license from a license server and not see it in the VCF Operations instance to which the license server is added, the license must not be assigned to any vCenter instance in your environment.

- Remove the license from all assets added to this VCF Operations instance. For more information, see Remove a License from a vCenter Instance.
- If your VCF Operations instance is in disconnected mode, report license usage. For more information, see Report License Usage and Update Licenses in Disconnected Mode.

To complete the license removal, you must update your licenses in VCF Operations. After you remove the license from VCF Operations, you can add the license to another VCF Operations instance and license server, or you can delete it.

1. Log in to the VCF Business Services console by using your Broadcom Support Portal credentials.
2. Select the Site ID to which you want to register this instance, and click **Next**.

   If you have only one site, it is selected by default.
3. Next to the license you want to remove, click the vertical ellipsis, and click **Remove License from License Server**.

   A dialog box appears.

4. Click **Remove**.
5. If your VCF Operations registration is in connected mode, update your licenses in the VCF Operations instance:
   a) In the VCF Operations instance, navigate to **Licensing** > **Licenses & Registration**.
   b) Click **Update Licenses**.
6. If your VCF Operations registration is in disconnected mode, manually upload the license file in VCF Operations:
   a) In the VCF Business Services console, in the Download the License File dialog box, click **Download**, and save the license file.
   b) Click **Finish**.
   c) In the VCF Operations instance, navigate to **Licensing** > **Licenses & Registration**.
   d) Click **Import License File**.
   e) In the **Import License File** dialog box, click **Browse**, select, and import the license file.
   f) Click **Complete**.

Delete a License

You can delete a split license from the VCF Business Services console console if you no longer need it. You cannot delete a split license if some of its capacity is assigned to a vCenter instance or another asset.

- Verify that the license is not assigned to a vCenter instance or another asset.
- Verify that the license is not added to a license server. For information on how to remove a license from a license server, see Remove a License from a License Server and the Respective VCF Operations Instance.

1. In the VCF Business Services console, navigate to **Licensing** > **Licenses**.
2. Next to the split license, click the vertical ellipsis, and click **Delete**.
3. In the dialog box which appears, click **Delete**.

The license is deleted and its capacity is merged with the default license capacity. In connected mode, after the new license file is automatically imported in the VCF Operations instance, the license that you deleted from the VCF Business Services console is automatically deleted from VCF Operations. In disconnected mode, you must manually download the license file from the VCF Business Services console and import it in VCF Operations. The license is then deleted automatically from VCF Operations.

Add a License to VCF Operations and License Server

To license your environment, you must first add licenses to the license server and VCF Operations instance that you registered with the VCF Business Services console. You cannot add one license to more than one license servers.

- Verify that you have a valid subscription for VCF or vSphere Foundation.
- Verify that your VCF Operations instance is registered with the VCF Business Services console.

After you add licenses to your license server, you can assign them to the vCenter instances.

A license cannot be added to more than one license server at a time.

1. In the VCF Business Services console, navigate to **Licensing** > **Licenses**.
2. Click **Add to VCF Operations**.

   The **Add Licenses to a VCF Operations Instance** wizard appears.
3. Select the licenses that you want to add, and click **Next**.
4. Select the VCF Operations instance to which you want to add the licenses, and click **Next**.
5. Select the license server to which you want to add the licenses, and click **Next**.
6. View the summary, and click **Save and Next**.
7. If your registration is in disconnected mode, download the license file, and click **Finish**.

   Licensing changes are communicated to VCF Operations by downloading a new license file from the VMware Business Services console into VCF Operations. VCF Operations is unaware of any changes made in the VCF

Business Services console until a new license file is downloaded, and continues to reflect the information in the last license file it received.

8. To complete the process and have your licenses ready to assign to assets, update your licenses in the VCF Operations instance.

Option	Description
Registration in connected mode	Navigate to **Licensing** > **Licenses & Registration**, and click **Update Licenses**. **Note:** You can automate this process by setting Update Licenses Mode to automated. With this setting, the new license file is downloaded automatically within 24 hours. You can still click **Update Licenses** to immediately download the license file.
Registration in disconnected mode	1. Navigate to **LicensingLicenses & Registration**. 2. Click **Import License File.** 3. In the **Import License File** pop-up window, click **Browse**. 4. Upload the license file, and click **Complete**.

The licenses are added to the VCF Operations instance. To view the license list in VCF Operations, navigate to **Licensing** > **Licenses**, and select the **Version 9+** tab.

Assign a Primary License to a vCenter Instance

You assign a primary license to a vCenter instance. Your other assets that are connected to that vCenter instance, including ESX hosts, are then licensed automatically. After you assign a primary license, you can also assign add-on licenses to license the assets that use add-on license capacity.

- Verify that you have the **Licensing** > **Manage**permission assigned in the VCF Operations instance that you use for license management. The Administrator role has this permission assigned by default.
- Verify that the VCF Operations instance is registered with the VCF Business Services console.
- Verify that at least one primary license is added to this VCF Operations instance.
- If you want to assign a license to a standalone vCenter instance, verify that the vCenter instance is integrated with the VCF Operations instance registered for license management. For more information about adding a vCenter instance to VCF Operations, see Configuring a vCenter Account in VCF Operations.
- If you want to assign a license to a vCenter instance that is part of a VCF domain, verify that the VCF Instance is integrated with the VCF Operations instance registered for license management. For more information about adding a VCF instance to VCF Operations, see Configuring a Standalone VCF Account in VCF Operations.

  Note:

  After you integrate a vCenter instance or a VCF Instance with VCF Operations for license management, it takes up to 20 minutes before the vCenter instance appears on the **Licenses** tab of VCF Operations.

- Verify that you have the required privileges in the vSphere Client to add a license to the vCenter instance.
  - Membership in the **LicenseService.Administrators** Single Sign-On group.
  - **Global > Licenses**
  - Either **Certificate Management** > **Create/ Delete(below Admins Priv)**, or **Certificate Management** > **Create/ Delete(below Admins Priv)**.
- Verify that the license has enough capacity to license all ESX hosts of versions 8.x, 9.0 and 9.1 that are added to the vCenter instance. For more information, see Mixed-Version Environments.

The vCenter instance can be standalone or part of a VCF Instance. You cannot assign a license to a vCenter instance if it requires more license capacity than the available capacity of the license you want to assign. If you try to assign the license to a vCenter instance which requires more capacity than the capacity available in the license, the assignment fails. If you try to add a host to a licensed vCenter instance, and the cores of the host are more than the available capacity left

in the license assigned to the vCenter instance, the host is added in evaluation mode. If the host has exceeded the 90-day evaluation period, you cannot add it to the vCenter instance unless you increase the available capacity of the license.

> **Note:**
>
> If the host is in evaluation mode, it is added to the vCenter instance even if the license capacity is not sufficient. During the evaluation period of the host, you must add more capacity to the license to license the host. Alternatively, you can assign an override license to the host. If the host is not licensed when the evaluation period expires, it is disconnected from the vCenter instance. The virtual machines running on the disconnected host remain intact. When there is enough available capacity in the license assigned to the vCenter instance, all disconnected hosts reconnect to the vCenter system.

For example, you add a license with 1000 cores allocated capacity to the VCF Operations instance. You then add 4 vCenter instances to be managed by this VCF Operations instance. The aggregate license consumption of the 4 vCenter instances requires 1500 cores. This means that you cannot assign the license to all of them, as the aggregate consumption across all vCenter instances cannot exceed 1000 cores. The aggregate consumption includes the total number of cores of all pre-version 9 and version 9+ hosts adeed to the vCenter instance, and excludes any license capacity used by ESX hosts licensed with override licenses.

For more information about calculating the license capacity you need for your environment, see Broadcom knowledge base article 95927.

When a vCenter license expires, your hosts are disconnected from the vCenter instance even if they have override licenses assigned to them, and you cannot start any workload operations. This is valid for both standalone vCenter instances and vCenter instances that are part of a VCF Instance. After you assign a valid license to the vCenter instance, the hosts are automatically reconnected to the vCenter instance unless you added them to another vCenter instance in the meantime.

If the license assignment to a selected vCenter instance in VCF Operations 9.1 fails with an error message, check the network connectivity between the vCenter instance and VCF Operations. For more information, see Broadcom Knowledge base article 424533.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, click **Licensing** > **Licenses & Registration**.
3. Select the **Version 9+** tab.
4. From the **vCenter Systems** table, select one or more vCenter instances to which you want to assign a license.

   The table contains only vCenter instances of version 9+ managed by the current VCF Operations instance. If a vCenter instance is added to the VCF Operations instance for monitoring only, it does not appear in this table.
5. Click **Assign Primary License**.

   A list with the primary licenses added to VCF Operations appears. The licenses which you cannot assign appear at the bottom of the list, and the column **Assignable** shows No.
6. Select a license to assign, and click **Assign**.

   > **Note:**
   >
   > You can assign only one primary license to a vCenter instance. If you want to replace the primary license with another one, see Remove a License from a vCenter Instance.

In the **Primary Licenses** column of the vCenter Systems table, you can view the licenses that you assigned to the vCenter instance. If the assignment failed for a vCenter instance, and you want to view the details of the task, refer to the tooltip message.

Assign an Add-on License to a vCenter Instance

Add-on licenses are available for VCF and vSphere Foundation products. You can assign add-on licenses only after you assign a primary license to a vCenter instance.

- Verify that you have the **Manage Licenses** permission assigned.
- Verify that your VCF Operations instance is registered with the VCF Business Services console.
- Verify that add-on licenses are added to the VCF Operations instance.
- Verify that the vCenter instance to which you want to assign a license is integrated with the VCF Operations instance. If the vCenter instance is part of a VCF domain, verify that the VCF Instance is integrated with the VCF Operations instance. For more information about adding vCenter instances and VCF Instances to VCF Operations, see Configuring a vCenter Account in VCF Operations, or Configuring a Standalone VCF Account in VCF Operations.
- Verify that you have the required privileges in the vSphere Client to add a license to the vCenter instance.
  - Membership in the **LicenseService.Administrators** Single Sign-On group.
  - **Global > Licenses**
  - Either **Certificate Management** > **Create/ Delete(below Admins Priv)**, or **Certificate Management** > **Create/ Delete(below Admins Priv)**.

You cannot assign a license to a vCenter instance if that instance requires more license capacity than the available license capacity of the license you want to assign. To be able to use an add-on license, you must first add a primary license to your VCF Operations instance, and assign it to the vCenter instances.

With your VCF subscription, in addition to your VMware Cloud Foundation (cores) license you receive a vSAN (TiBs) license. Starting November 3, 2025, depending on your VCF subscription, you might also receive a VMware Private AI Foundation with NVIDIA (cores) license.

If the license assignment to a selected vCenter instance in VCF Operations 9.1 fails with an error message, check the network connectivity between the vCenter instance and VCF Operations. For more information, see Broadcom Knowledge base article 424533.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, click **Licensing** > **Licenses & Registration**.
3. Select the **Version 9+** tab.
4. From the vCenter Systems table, select one or more vCenter instances to which you want to assign a license.

   The table contains only vCenter instances of version 9+ managed by the current VCF Operations instance. If a vCenter instance is added to the VCF Operations instance for monitoring only, it does not appear in this table.

   For VMware Private AI Foundation with NVIDIA, assigning a VMware Private AI Foundation with NVIDIA license to the management domain is required to activate the guided deployment UI in the vSphere Client. The license capacity is allocated only to the GPU-enabled workload domains and not to the management domain.
5. Click **Assign Add-on License**.
6. From the drop-down menu, select the product.

   A list with the add-on licenses of the type you selected that are added to VCF Operations appears. You cannot assign licenses which do not have the required capacity to license the selected vCenter instances. These licenses appear at the bottom of the list, and the column Assignable shows No.
7. Select a license to assign, and click **Assign**.

The add-on licenses are assigned to the vCenter instance. To view them, click the vCenter instance in the vCenter Systems table.

If the assignment failed for a vCenter instance, and you want to view the details of the task, click View Details in the banner above the table and hover over the information icon in the Status column.

What is an Override License

When you assign a license to a vCenter instance, it is automatically assigned to all assets of that type in your environment that are connected to this vCenter instance. If you want to assign a different license to an asset, you can assign it directly as an override license. Override licenses are available only for assets which use license capacity, such as ESX hosts and vSAN clusters.

You cannot assign an override license to a standalone asset, the asset must be added to a vCenter instance before you can assign the override license. The override license and the vCenter license can be from the same or from different license servers, if the license servers are added to the same VCF Operations instance.

Note:

You cannot assign override licenses to ESX hosts of versions earlier than 9.0.

When you remove a host from the vCenter instance, the license capacity it uses is added back to the license assigned to the host. If the host is licensed with an override license, the license capacity is added back to the override license. If the host is licensed with a vCenter license, the license capacity is added back to the vCenter license.

If the vCenter license expires, you can assign an override license to an ESX host but you must also assign a valid license to the vCenter instance. If you do not assign a valid license to the vCenter instance, you cannot use the management capabilities of the vCenter instance and you can only manage the host as a standalone host.

Evaluation Mode and Override Licenses

When you add an ESX host of version 9.0 or later to a vCenter instance of version 9.1, the vCenter license is assigned to the host automatically if the license capacity is sufficient to license the host. If the license capacity is insufficient, the host is added in evaluation mode. The evaluation period is 90 days, during that time you can either add more capacity to the vCenter license, or assign an override license to the host.

Note:

If the evaluation mode for the host has expired before you try to add it to the vCenter instance, you cannot assign an override license to it. If the vCenter license capacity is insufficient, to add the host, you must increase the license capacity.

If you add hosts to a vCenter instance when the vCenter instance is in evaluation mode, you must assign a valid license to the vCenter instance before its evaluation period expires. If you do not assign a valid license to the vCenter instance, when the evaluation period expires, you cannot use the vCenter instance and all hosts are disconnected from it.

License Compatibility

The override and vCenter license must meet compatibility criteria. Compatibility is checked at the time when you assign a license. If the compatibility check fails, the license assignment also fails.

When you assign an override license, a compatibility check is performed between the vCenter license and the override license.

- If a VCF license is assigned to the vCenter instance, you cannot assign a VVF license as an override license.
- If a VVF license is assigned to the vCenter instance, you can assign a VCF license as an override license.

When you assign a vCenter license, a compatibility check is performed between the vCenter license and any override licenses assigned directly to hosts or clusters.

- If a VCF license is assigned as an override license, you can assign a VVF license to the vCenter instance.
- If a VVF license is assigned as an override license, you cannot assign a VCF license to the vCenter.

Note:

VCF and VCF Edge are equivalent for the purpose of compatibility.

vSAN licenses and vSAN Edge licenses are not compatible.

Assign an Override License to an ESX Host in VCF Operations

Override licenses are available for assets which use license capacity, such as ESX hosts and vSAN clusters.

- Verify that the host is added to a vCenter instance.
- Verify that the override license has sufficient license capacity to license the host.

You cannot assign an override license to a standalone asset, the ESX host must be added to a licensed vCenter instance before you can assign the override license.

Note:

You cannot assign override licenses to ESX hosts of versions earlier than 9.0.

The override and vCenter license must meet compatibility criteria. Compatibility is checked at the time when you assign a license. If the compatibility check fails, the license assignment is unsuccessful. For more information about override licenses, see What is an Override License.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, click **Licensing** > **Licenses & Registration**.
3. Select the **Version 9+** tab.
4. From the vCenter Systems table, click the vCenter instance to which the ESX host is added.

   The table contains only vCenter instances of version 9+ managed by the current VCF Operations instance. If a vCenter instance is added to the VCF Operations instance for monitoring only, it does not appear in this table.
5. On the page with the vCenter license details, navigate to **Assets**, and click **ESX Hosts**.
6. Select the check box next to the host to which you want to assign an override license, and click **Assign Override License**.
7. On the Assign Override License page, select the license that you want to assign to the host, and click **Assign**.

The license is assigned to the ESX host.

Replace an Override License with a vCenter License

You can replace an override license with the license assigned to the vCenter instance at any time if the vCenter license has enough available capacity to license the asset.

- Verify that the vCenter license has enough available license capacity to license the asset.

  Note:

  If a license is assigned as an override license, and you later assign it to the vCenter instance, it is considered a vCenter license and replaces the override assignment.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, click **Licensing** > **Licenses & Registration**.
3. Select the **Version 9+** tab.
4. Identify the vCenter instance that contains the asset to which you want to assign an override license.
5. From the vCenter Systems table, click the vCenter instance that contains the asset.

   The table contains only vCenter instances of version 9+ managed by the current VCF Operations instance. If a vCenter instance is added to the VCF Operations instance for monitoring only, it does not appear in this table.
6. On the page with the vCenter license details, navigate to **Assets**.
7. Assign a vCenter license to an ESX host.
   a) Under **Assets**, click **ESX Hosts**.
   b) Select the check box next to the hosts to which you want to assign an override license, and click **Switch to vCenter License**.

      A dialog box appears.
   c) Click **Replace**.

      The ESX hosts are licensed with the license assigned to the vCenter instance. The License Source column shows vCenter.

8. Assign a vCenter license to a vSAN cluster.
   a) Under **Assets**, click **vSAN Clusters**.
   b) Select the check box next to the vSAN clusters to which you want to assign an override license, and click **Switch to vCenter License**.

      A dialog box appears.
   c) Click **Replace**.

      The vSAN clusters are licensed with the license assigned to the vCenter instance. The License Source column shows vCenter.

Reclaim Used Capacity

When you delete an asset, your license capacity is automatically reclaimed and available to assign to a new asset. If a connectivity issue between the license server and the vCenter instance occurs, the license capacity might not be automatically added back to the license. If this happens, you can manually reclaim the used capacity.

- Verify that the asset is removed from the vCenter instance.
- If you want to reclaim capacity used by a vCenter instance, verify that the vCenter instance is removed from the current VCF Operations instance.

To reclaim the license capacity used by a vCenter instance and the assets connected to it, you must first remove the vCenter instance from the VCF Operations instance. If you want to continue monitoring the vCenter instance from the VCF Operations instance, you can remove the vCenter instance only from license management.

When you remove the vCenter account from VCF Operations, the license capacity used by assets connected to the vCenter instance is added back to the license, and the vCenter instance no longer communicates with the license server.

> **Note:**
>
> To avoid service interruption for the vCenter instance, you must add it to a VCF Operations instance, and assign a license to it.

In some cases, used capacity might not be automatically reclaimed after you remove assets from your environment. For example, if connectivity between the license server and the vCenter instance is interrupted at the time when you remove the vCenter instance from VCF Operations, or when you remove an ESX host from vCenter . In such cases you must manually reclaim the used capacity. You can reclaim used capacity only for assets that are no longer in the inventory, for example, hosts or clusters that are removed from the vCenter instance, or vCenter instanced that are removed from the VCF Operations instance.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, click **Licensing** > **Licenses & Registration**.
3. In the Registration and License Server Status card, click **Manage** > **License Servers**.

   The **VCF Operations Registration** page appears.
4. On the **License Servers** tab, click the license server which contains the license that is assigned to the removed asset.
5. Under Licensed Assets, select the asset for which you want to reclaim the capacity, and click **Reclaim Used Capacity**.

   A dialog box appears.
6. Click **Reclaim Used Capacity**.

The license capacity used by the removed asset is added back to the license as available capacity.

Updating Licenses and Viewing the License Usage File

You must update your licenses in VCF Operations to verify that a usage file has been submitted, or when you make a license-related change such as subscription capacity changes, renewals, or splitting licenses and changing their capacity. When you update licenses, you do not need to reassign licenses to the assets.

If you do not make any license-related changes in the VCF Business Services console, you must still update your licenses at least once every 6 months (180 days). If license usage data is not submitted within the required reporting time-frame, and you do not update your licenses, your licenses are treated as expired, your hosts are disconnected from the vCenter instance, and you cannot start any workload operations.

How to view the detailed contents of your usage file

License usage files contain information on the capacity of each license used by a VCF fleet. VCF Operations creates the license usage files and sends them to the VCF Business Services console.

You can view your license usage in a ready-to-read usage file that is decompressed and decoded, or you can view the raw usage file that is created in your VCF Operations instance. The raw usage file is a JSON Web Signed (JWS) file compressed with gZip.

Note:

- If your VCF Operations instance is in connected mode, usage files are automatically generated and submitted daily and you can only view the ready-to-read usage file.
- If your VCF Operations instance is in disconnected mode, you must manually download and submit usage files on regular intervals.

The contents of the usage files are identical in connected and disconnected mode.

View a ready-to-read usage file	1. In the VCF Operations instance, navigate to **Licensing** > **Registration**. 2. Under **File History**, download the usage file that you want to view, and open it.
View the raw usage file in a human-readable format	1. In the VCF Operations instance, navigate to **Licensing** > **Registration**, and click **Generate Usage File**. 2. Use decompression software that supports gZip decompression to decompress the content of the usage file. 3. Use a text editor to open the decompressed file, and copy the content. 4. Paste the content of the usage file into a JWT Decoding tool. After the content is decoded, you can view the collected license usage information.

License Usage File Content

Property	Applies to version	Description
usage_report_generated_time	9.0 and later	The date and time when the usage report is generated.
acknowledged_claims	9.1 and later	This property is for reserved licenses. If you have no reserved licenses, the property is *NULL*.
additional_data	9.1 and later	Additional records include the count of objects instead of just aggregate core and TiB counts. For more information, see the table Table 982: Additional Data in the Usage File.
usage_report_id	9.0 and later	The unique ID of this usage report.
asset_id	9.0 and later	The unique ID of the VCF Operations instance where data is collected.
asset_status	9.0 and later	*Active*.
xr2	9.0 and later	A unique opaque fingerprint of the VCF Operations instance where data is collected; used in conjunction with the asset_id to link the usage data to the registered VCF Operations instance. No identifiable environment details can be derived from the fingerprint.

Property	Applies to version	Description
jws_child_asset_usages	9.1 and later	For each license server that is available for this VCF Operations instance, a unique usage report is included that reflects changes to the license usage. This content is encoded. You can decode it by using any JWT token decoder.
product_version	9.0 and later	• *9.0* when the license is version 9+. • *1.0* when a pre-version 9 license key is used.
product_name	9.0 and later	Product display name.
Allocation_id	9.0 and later	License ID for the version 9+ license.
license_key_id	9.0 and later	License key for a pre-version 9 license.
usage_start_time	9.0 and later	Start time of the usage period.
usage_end_time	9.0 and later	End time of the usage period.
granularity	9.0 and later	Reflects the cadence in which changes are recorded. Always *minutes*.
uom	9.0 and later	TiB or core for version 9+ licenses.
quantity	9.0 and later	License usage quantity for this usage period.
total_quantity	9.0 and later	License capacity, included for pre-version 9 licenses only.
pre_v9_quantity	9.1 and later	License usage of a version 9+ license by a pre-version 9 host.
unassigned_licenses	9.0 and later	List of version 9 licenses (allocation_id) that are added to VCF Operations and are not used.
license_usage_anomaly_details	9.0 and later	Indicates cases where usage data was corrupted, possibly as a result of a corrupt database, corrupt software update, or related circumstances.
key_rotation_details	9.1 and later	Used to coordinate public key rotations.
manually_freed_capacity.total_freed_capacity	9.1 and later	Indicates the amount of used license capacity that is manually reclaimed from the license server and for which license.
recycled_events.event_type	9.1 and later	Indicates the method in which license usage data was reclaimed from the license server and later re-used.
recycled_events.asset_type	9.1 and later	Indicates the type of asset that was released from the license server used capacity. Example: *host*.
recycled_events.RecycleStats.assets_recycled	9.1 and later	The count of licensed assets for which license capacity was reclaimed and were later relicensed.
recycled_events.RecycleStats.max_recycle_frequency	9.1 and later	The highest frequency in which a licensed asset received a new license after having the license reclaimed.
recycled_events.avg_recycle_frequency	9.1 and later	The average frequency in which licensed assets received a new license after having it removed.

Additional records in the usage file include the count of objects instead of only aggregate core and TiB counts. The *additional_data* object from the license usage file contains several properties.

Table 982: Additional Data in the Usage File

Property	Applies to version	Description
id	9.1 and later	Unique identifier for the VCF Operations instance.
vcenters	9.1 and later	List of vCenter system objects included in the report.
vc_guid	9.1 and later	The unique Global Unique Identifier (GUID) for the vCenter instance.
vc_version	9.1 and later	The software version of the vCenter server.
host_info.total	9.1 and later	Total count of ESX hosts managed by this vCenter instance.
host_versions.version	9.1 and later	The software version of the ESX hosts.

Property	Applies to version	Description
host_versions.total_count	9.1 and later	Total count of hosts of that specific version.
cluster_info.total	9.1 and later	Total number of clusters managed by the vCenter instance.
cluster_info.vks_clusters	9.1 and later	Number of VMware Kubernetes Service (vKS) clusters.
cluster_info.vsan_enabled_clusters	9.1 and later	Number of clusters with vSAN storage enabled.
cluster_info.vsan_data_stores	9.1 and later	Total number of vSAN datastores across the clusters.

The license usage file exclusively contains this specific information and does not collect personal data or customer data.

Update Licenses in Connected Mode

After any change to your licenses in the VCF Business Services console, to reflect the changes in your on-premise environment, your licenses must be updated in the VCF Operations instance. In connected mode, the usage file is submitted automatically on a daily basis. If your licenses update mode is set to manual, you must still update your licenses in the VCF Operations instance at least once every 180 days to verify that the usage file has been submitted correctly, and to keep your environment licensed.

You must update your licenses at least once every 6 months (180 days). If license usage data is not submitted within the required reporting time-frame, and you do not update your licenses, your licenses are treated as expired, your hosts are disconnected from the vCenter instance, and you cannot start any workload operations. You can update your licenses at any time to extend your usage reporting refresh period.

> **Note:**
>
> If your licenses update mode is set to automated, your licenses are updated automatically once every 24 hours and after each change you make to the licenses related to the VCF Operations instance. For more information about switching to automated licenses update mode, see Change License Updates Mode.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, click **Licensing** > **Licenses & Registration**.
3. Optional: To verify that your license usage file was submitted within the last 24 hours, click **Manage** > **Registration Details**, and check the **Last Usage Report Generated** date**.**
4. Click **Update Licenses**.

Your VCF Operations instance is updated with the latest license information available from the VCF Business Services console.

Change License Updates Mode

If your VCF Operation instance operates in connected mode, you can choose from an automated and a manual update of licenses. The default license updates mode is automated.

The automated update of licenses happens once every 24 hours and after each change you make to the licenses related to the VCF Operations instance. The manual update requires that you click a button to update the licenses after each change and at least once every 180 days. License usage reporting is automated in both the manual and the automated license update mode because your environment is in connected mode.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, click **Licensing** > **Licenses & Registration**.
3. In the Registration and License Server Status card, click **Manage** > **Registration Details**.

   The **VCF Operations Registration** page appears.
4. On the **Registration Details** tab, switch the mode for updating licenses.
   - If you want to switch from automated to manual mode, next to Update Licenses Mode, click Switch to Manual.

- If you want to switch from manual to automated mode, next to Update Licenses Mode, click Switch to Automated.

Report License Usage and Update Licenses in Disconnected Mode

When a VCF Operations instance is in disconnected mode, you must manually exchange the usage file and the license file between the VCF Operations instance and the VCF Business Services console.

- Verify that your VCF Operations instance is registered in the VCF Business Services console.

You must update your licenses and submit a license usage file at least once every 6 months (180 days). If license usage data is not submitted within the required reporting time-frame, and you do not update your licenses, your licenses are treated as expired, your hosts are disconnected from the vCenter instance, and you cannot start any workload operations.

You can generate a new license file without submitting a usage file as long as it is within your usage reporting due date. As a best practice, always submit a usage file before you download a new license file.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, click **Licensing** > **Licenses & Registration**.
3. Click **Generate Usage File**, and save the usage file.
4. In the VCF Business Services console, navigate to **Licensing** > **VCF Operations Registrations**.
5. Click the VCF Operations instance for which you want to update licenses.
6. In the **VCF Operations Details** page, click **Actions** > **Upload Usage File**.
7. In the **Upload Usage File** wizard, upload the usage file, and click **Upload and Next**.

   A license file is generated.
8. Click **Download**, and save the license file.
9. Click **Finish**.
10. In the VCF Operations instance, navigate to **Licensing** > **Licenses & Registration**.
11. In the Registration and License Server Status card, click **Manage** > **Registration Details**.
12. Click **Import License File**.

    The **Import License File** dialog box appears.
13. Click **Browse**, upload the license file, and click **Import**.
14. When the license file is imported, click **Complete**.

    A dialog box appears. You can view the licenses and the license servers that are added to this VCF Operation instance.

Remove a License from a vCenter Instance

To remove a primary license from a vCenter instance, you must replace it with another primary license.

- Verify that you have the **Manage Licenses** permission assigned.
- Verify that licenses are added to this VCF Operations instance.
- Verify that the license you want to assign to the vCenter instance has sufficient capacity.

If you no longer want to keep the license assigned to a vCenter instance, you can replace it with another primary license. The used license capacity of the license you replace is freed up and you can assign it to other assets.

When you replace a primary license assigned to a vCenter instance with another primary license, you can choose whether to update override license assignments. You can replace all override licenses assigned to the hosts in that vCenter instance with the new primary license or you can keep the override licenses assigned to the hosts. If you want to keep the override license assignments, license compatibility is checked. For more information about the compatibility matrix, see What is an Override License.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, click **Licensing** > **Licenses & Registration**.
3. Select the **Version 9+** tab.
4. From the vCenter Systems table, click the vCenter instance.

   The table contains only vCenter instances of version 9+ managed by the current VCF Operations instance. If a vCenter instance is added to the VCF Operations instance for monitoring only, it does not appear in this table.
5. In the page with the vCenter license details, navigate to **Licenses**, and click **Replace Primary License**.

   A list with the licenses added to VCF Operations appears. The list also includes the currently assigned license.
6. Select a license to assign, and click **Assign**.

The new license is assigned to the vCenter instance, and you can view it in the Licenses section.

Remove a vCenter Account from VCF Operations for License Management

If you want to move a vCenter instance to a different VCF Operations instance for license management, but continue to monitor the vCenter instance in the same VCF Operations instance, you can deactivate only license management for the vCenter instance.

Verify that you have the required privileges.

A vCenter instance can be added to a VCF Operations instance just for monitoring or for monitoring, license management, and vSphere Client Plug-in management.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Home** and in the left navigation pane, click **Administration** > **Integrations**.
3. Next to the vCenter instance that you want to remove for license management, click the vertical ellipsis, and click **Edit**.

   The Cloud Account Information page for the vCenter instance appears.
4. At the bottom of the page, click **Manage Integration**.

   The Manage Integration dialog box appears.
5. To remove the license management and vSphere Client plug-in management for the vCenter instance, switch the toggle button **License and vSphere Client plug-in management are active** to **Off**.
6. Click **Save**.

Delete a License Server

If you want to replace your license server with a new one, and you want to delete the first license server, you must delete it from the VCF Business Services console and the VCF Operations instance.

To keep your environment licensed, before you delete a license server, you must transfer the licenses added to it to another license server that is registered with the same VCF Operations instance. Alternatively, you can assign new licenses to all assets in your environment. The new licenses must be added to a license server that is registered with the same VCF Operations instance.

You must first delete the license server from the VCF Business Services console, and then you can delete it from the VCF Operations instance.

> **Note:**
>
> If there are licenses added to the license server, and you delete the license server, the licenses are removed. As a result, all assets licensed with these licenses become unlicensed. To avoid service interruption, assign new licenses to the assets.

1. Log in to the VCF Business Services console.
2. Navigate to **Licensing** > **VCF Operations Registrations**.
3. Click the VCF Operation instance which contains the license server you want to delete.
4. On the VCF Operation instance Details page, navigate to the **License Servers** table.
5. Next to the license server you want to delete, click the vertical ellipsis button (⋮), and click **Delete License Server**.

   A pop-up appears.
6. If there are no licenses added to the license server, in the **Delete License Server** pop-up, click **Delete**.
7. If there are licenses added to the license server, in the **Delete License Server** pop-up, select the check box, and click **Delete**.

   The licenses are removed. All assets licensed with these licenses become unlicensed. To avoid service interruption, assign new licenses to the assets.
8. If your VCF Operations instance is in disconnected mode, you must download the updated license file and import it in VCF Operations.
   a) From the **Delete License Server** pop-up, download the updated license file, and click **Finish**.
   b) Log in to the VCF Operations instance, and navigate to **Licensing** > **Licenses & Registration**.
   c) In the Registration and License Server Status card, click **Manage** > **Registration Details**.
   d) Click **Import License File**.

      The **Import License File** dialog box appears.
   e) Click **Browse**, upload the license file, and click **Import**.
   f) When the license file is imported, click **Complete**.

      A dialog box appears. You can view the licenses and the license servers that are added to this VCF Operation instance.
9. If your VCF Operations instance is in connected mode, update your licenses in the VCF Operations instance.
   a) In the **Delete License Server** pop-up, click **Finish**.
   b) Log in to VCF Operations.
   c) From the navigation bar at the top, click **Manage** and in the left navigation pane, click **Licensing** > **Licenses & Registration**.
   d) Click **Update Licenses**.
10. Delete the license server from the VCF Operations instance.
    a) In the left navigation pane, click **Licensing** > **Licenses & Registration**.
    b) In the Registration and License Server Status card, click **Manage** > **License Servers**.

       The **VCF Operations Registration** page appears.
    c) On the **License Servers** tab, click the license server, under License Servers, click the vertical ellipsis next to the license server.
    d) Click **Delete**.

       The Delete a License Server dialog box appears.
    e) Click **Delete**.

The license server is deleted. If you do not have other license servers added to the VCF Operations instance, you cannot assign licenses to the assets related to that VCF Operations instance. To use the VCF Operations instance for license management, you must have at least one license server added to it. For information on how to add a license server, see Deploy a License Server Version 9.1.0.

Deactivate a VCF Operations Instance

When you deactivate a VCF Operations instance from the VCF Business Services console, you can no longer add any licenses to that instance, and you cannot perform any license management operations.

- Verify that there are no licensed vCenter or VCF Instances managed by this VCF Operations instance.
- Delete all license servers from the VCF Operations instance. For more information, see Delete a License Server.
- If the VCF Operations instance is in disconnected mode, report license usage. For more information, see Report License Usage and Update Licenses in Disconnected Mode.

Note:

If you no longer have access to the VCF Operations instance, you can deactivate the instance without following the prerequisites listed above.

Once you deactivate a VCF Operations instance, if you want to reactivate it, the same licenses that are added to it before the deactivation might not be available.

Note:

When you deactivate a VCF Operations instance, all licenses servers that are added to this instance are automatically deleted.

1. In the VCF Business Services console, navigate to **Licensing** > **VCF Operations Registrations**.
2. Click the vertical ellipsis next to VCF Operations instance that you want to deactivate.
3. Click **Deactivate**.

   A dialog box appears.
4. If you do not have access to the VCF Operations instance and you want to deactivate it, select the check box **I no longer have access to the VCF Operations instance**.
5. Click **Deactivate**.

The VCF Operations instance is deactivated. You can view it in the list of VCF Operations Registrations with a status **Deactivated**.

Reactivate a VCF Operations Instance

You can reactivate a deactivated VCF Operations instance.

If you want to restore the same license configuration that was in use before the VCF Operations instance deactivation, add the same licenses to the instance that were added before the deactivation. If those licenses are not available, add new licenses to the instance.

1. In the VCF Business Services console, navigate to **Licensing** > **VCF Operations Registrations**.
2. Click the VCF Operations instance that you want to reactivate.

   The details and licenses page for the instance appears.
3. Click **Reactivate**.

   A dialog box appears.
4. In the **Reactivate VCF Operations** dialog box, click **Reactivate**.
5. Add licenses to the VCF Operations instance. For information on how to add licenses, see Add a License to VCF Operations and License Server.
6. Update your registration with the VCF Business Services console.

Option	Description
Connected Registration	1. In the VCF Business Services console, click the vertical ellipsis next to the VCF Operations instance. 2. Click **Generate Activation code**. 3. In the **Generate Activation Code** dialog box, copy the activation code, and click **Finish**. 4. In the VCF Operations instance, navigate to **Licensing** > **License and Registration**.

Option	Description
	5. In the Registration and License Server Status card, click **Manage** > **Registration Details**. The VCF Operations Registration page appears, and the Registration Details tab is selected. 6. Click **Actions** > **Enter Activation Code**. 7. In the **Enter Activation Code** dialog box, paste the activation code, and click **Activate**.
Disconnected Registration	1. In the VCF Operations instance, navigate to **Licensing** > **Licenses & Registration**, and click **Generate Usage File**. 2. In the VCF Business Services console, navigate to **Licensing** > **VCF Operations Registrations**, and click **Upload Usage File**. 3. In the **Upload Usage File** wizard, click **Browse** and upload the usage file that you generated in the VCF Operations instance. 4. Click **Upload and Next**. A license file is generated. 5. Download the license file, and click **Finish**. 6. In the VCF Operations instance, navigate to **Licensing** > **Licenses & Registration**, and click **Import License File**. 7. In the **Import License File** dialog box, click **Browse** and import the license file. 8. Click **Complete**.

Your VCF Operations instance is reactivated and you can use it for license management.

License Usage Analytics

License usage analytics are available both in the VCF Operations instance that you use for license management, and in the VCF Business Services console. You can access your license usage data at any time to check the aggregate license usage per product and per license.

With your subscription for a product, you also receive license keys for version 8.x of the product. You can decide which version of the product to deploy and license, but your total license usage must not exceed your purchased subscription capacity. You can use the usage analytics data to monitor license usage and change license capacity.

Note:

If at a certain point the used capacity is higher than the capacity allocated to the assigned licenses, you must allocate more capacity to the licenses. Before you add more capacity, you cannot assign the over used licenses to new vCenter instances. To the vCenter instances which are licensed with the over used licenses, you can only add new ESX hosts that are in evaluation mode.

View License Usage in the VCF Business Services console

In the VCF Business Services console, you can view aggregated data for your complete license usage for all your products and licenses in the unit of measure for the specific license. You can use the analytics data to monitor license usage and plan your license management.

License usage is determined based on the vCenter version.

- vCenter 9.1 and later. Version 9.0+ usage combines license usage by hosts of version earlier than 9.0 that use license keys, and hosts of version 9.0 and later that use license files. When hosts of versions earlier than 9.0 that use license keys are managed by vCenter instances of version 9.1 or later, the license capacity of these hosts is included in the version 9+ license usage.
- For vCenter 9.0. License usage is separated into license usage by hosts of version earlier than 9.0 that use license keys, and hosts of version 9.0 that use license files.
- For vCenter 8.x and earlier. License usage by hosts of version 8.x or earlier that use license keys.

The available usage tabs provide detailed breakdowns:

- Pre-Version 9 License Usage. Includes license key usage by pre-9.0 hosts managed by vCenter instances of version 9.0 and license key usage by hosts of version 8.x or earlier managed by vCenter instances of version 8.x.
- Version 9+ License Usage. Includes license usage by hosts managed by vCenter instances of version 9.1 and later.
- Product Aggregate Usage. Presents the total usage of all licenses for each product, derived from the **Version 9+ License Usage** tab.

You can also export and view the usage analytics data for product aggregate usage, version 9+ license usage, and pre-version 9 license usage. The export file is available in CSV and JSON formats.

1. Log in to the VCF Business Services console (vcf.broadcom.com).
2. Navigate to **License Management** > **Usage Analytics**.
3. View detailed license usage per product or per license.

Usage by Product

1. In the **Usage by Product** pane, click the product for which you want to view detailed license usage.
   The **Usage Details** page for the selected product appears.

   Note:

   The **Usage by Product** pane includes the top five products based on highest percentage of license usage. You can view the full product list on the **Product Aggregate Usage** tab.
2. To view usage analytics for a specific license for a product of version 9.0 or later, select the license from the **Select License** drop-down menu.
3. To view usage analytics for a specific license key for a product of version earlier than 9.0, select the license key from the **Select License Key** drop-down menu.
4. In the **Usage History** pane, you can view maximum license usage for a specific aggregation period. Use the filters to view maximum license usage in monthly or daily periods.

Product Aggregate Usage

1. Select the **Product Aggregate Usage** tab.
2. Click the product for which you want to view detailed license usage.
   The **Usage Details** page for the selected product appears.
3. To view usage analytics for a specific license for a product of version 9.0 or later, select the license from the **Select License** drop-down menu.
4. To view usage analytics for a specific license key for a product of version earlier than 9.0, select the license key from the **Select License Key** drop-down menu.
5. In the **Usage History** pane, you can view maximum license usage for a specific aggregation period. Use the filters to view maximum license usage in monthly or daily periods.

Version 9+ License Usage

1. Select the **Version 9+ License Usage** tab.
2. Click the license for which you want to view detailed license usage.
   The **Usage Details** page for the selected license appears.
3. In the **Usage History** pane, you can view maximum license usage for a specific aggregation period. Use the filters to view maximum license usage in monthly or daily periods.

Pre-Version 9 License Usage

1. Select the **Pre-Version 9 License Usage** tab.
2. Click the license key for which you want to view detailed license usage.
   The **Usage Details** page for the selected license appears.
3. In the **Usage History** pane, you can view maximum license usage for a specific aggregation period. Use the filters to view maximum license usage in monthly or daily periods.

View License Usage Per Product in VCF Operations

In your VCF Operations instance, you can view aggregated data for your license usage for products and for specific licenses, based on the licenses added to this VCF Operations instance. You can use the analytics data to monitor license usage and plan your license management.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, click **Licensing** > **Usage Analytics**.
3. In the **Usage by Product** pane, click the product for which you want to view detailed license usage.

   Note:

   The pane includes the top five products based on the highest percentage of license usage. You can view the full product list on the **Product Aggregate Usage** tab.

   The usage details page for the selected product appears.
4. If you want to view usage analytics for a specific license server, use the **Select License Server** drop-down menu.
5. If you want to view usage analytics for a specific license, use the **Select License** drop-down menu.
6. If you want to view usage analytics for a specific vCenter instance, use the **Select vCenter** drop-down menu.

   The list of vCenter instances includes only the instances to which you have assigned licenses from this VCF Operations instance. If you select a specific license to view data for, the list of vCenter instances contains only the vCenter instances to which this license is assigned.
7. In the **Historical Usage** pane, you can view maximum license usage for a specific aggregation period.

   Use the filters to view maximum license usage in monthly or daily periods.

License Management for Cloud Services Providers and Hyperscalers

You license your VCF environment by using a VCF Operations instance and the VMware Cloud Foundation Business Services console (vcf.broadcom.com).

Overview

- To license your environment, you must have entitlements to the VCF products.
- After a commit contract is signed and available in the Broadcom Support Portal, you must create licenses against the contract serial number and the product in the contract.
- If you and the end customer meet certain requirements, end customers can share licenses for portability with you. For more information, see License Sharing.
- To license your environment, you must have an installed VCF Operations instance and a license server, register them in the VCF Business Services console, and add licenses with available capacity to a license server. You can register VCF Operations to the VCF Business Services console either in connected or in disconnected mode. Connected registration is recommended for cloud services providers and hyperscalers.
- You must update your licenses at least once every six months. For more information, see Updating Licenses and Viewing the License Usage File.
- To collect the relevant product usage, you must deploy a VCF usage meter instance. VCF usage meter collects the licensing data and usage data (for more information, see the guides in the Broadcom Partner Portal Page for VMware Cloud Service Providers under VCSP Portfolio) in the metered environments and communicates the data to the cloud. The data is converted to core usage per hour for each license.
- You must register the VCF usage meter instance in the VCF Business Services console. The product usage data is available in the VCF Business Services console for partners to download and review.

License Management Between Tenants and Sites

You can manage licenses in all the tenants of the sites you belong to. However, you cannot transfer licenses between tenants. You must register the VCF Operations instance in the same tenant as the licenses you want to add. If you register the VCF Operations in a wrong tenant, you can deactivate the VCF Operations instance and restart the registration in the correct tenant. If you have added licenses, the licenses must first be removed before deactivating the VCF Operations instance.

Licensing Attributes for Providers

On the **Licensing** tab of the VCF Business Services console, you can see the attributes of the licenses you manage.

Licensing Attributes

Component	Description
Commit Capacity	The committed capacity in the signed commit contract. Can be in a number of cores or TiBs.
Total Licenses	Total number of the created licenses.
Consumed Capacity	The consumed capacity contains data for both non-running and running workloads. If VCF usage meter manages the environment on a running workload and the consumed capacity exceeds the commit capacity, you might get billed for over use.

High-Level Provider Licensing Workflow

1. Install a VCF Operations instance version 9.1, or upgrade your existing Aria Operations instance to VCF Operations 9.1.
2. Register your VCF Operations instance with the VCF Business Services console by using one of the two modes.
   - Disconnected mode - no Internet connection is required. You must manually exchange registration, verification, and license files between the VCF Operations instance and the VCF Business Services console.
   - Connected mode - Internet connection is required. You use an activation code from the VCF Business Services console to register your VCF Operations instance and the license servers.
3. Add licenses to the license servers from the VCF Business Services console.
4. Assign licenses to vCenter instances from VCF Operations.
5. Deploy or upgrade a VCF usage meter appliance to version 9.1.

Create Licenses as a Provider

After you purchase a contract serial number and a product, you must create licenses against a contract serial number combination.

Verify that you are in the correct tenant.

Depending on the type of your contract, you can exceed the commit capacity for the created licenses. The capacity of the licenses must be between 1 and 100 000 cores.

1. Log in to the VCF Business Services console.
2. Navigate to **License Management** > **Licenses**.
3. Click **Manage Licenses**.
4. From the **All Serial Numbers and Products**, select the serial number and the product you want to create the new license against.

   You can see all of the licenses previously created for the selected serial number and product with high-level details for commit contract capacity, the maximum capacity, total number of licenses created, and the total allocated capacity. You can also see the expiration date and consumed capacity of each license individually.
5. Click **Create**.
6. Enter a name and capacity for the license and click **Next**.
7. View the summary and click **Finish**.

Register VCF Operations as a Provider

If the VCF Operations instance is provider managed, you should register it in connected mode.

Verify that you register the VCF Operations in the correct site and tenant.

The registered VCF Operations instance can be

- fully provider managed
- both customer and provider managed (applicable to white label customers)

If the VCF Operations is fully provider managed, you register the VCF Operations instance in connected or disconnected mode and manage all the licenses for the end customers within the tenant.

If both a customer and provider manage the VCF Operations instance, you must start the registration in disconnected mode and then pass the activation code to the end customer. For more information, see Register VCF Operations and Your License Server in Connected Mode and Register VCF Operations and Your License Server in Disconnected Mode.

Note:

When you add licenses in the VCF Business Services console, you can only select from existing licenses. Add only licenses created in the same tenant as the one you register the VCF Operations instance in.

Restart a VCF Operations Registration

If you register the VCF Operations in the wrong site or tenant, you can restart the VCF Operations registration.

Before you restart the VCF Operations registration, you must remove any licenses added to the VCF Operations instance, delete the VCF Operations instance, and then re-register in the correct tenant. For more information, see Delete a License.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, click **Licensing** > **Licenses & Registration**.
3. In the **Registration and License Server Status** pane, navigate to **Manage** > **Registration details**.

   The **VCF Operations Registration** page appears.
4. On the **Registration Details** tab, navigate to **Actions** > **Restart Registration**.

Move a Contract or a Serial Number Between Site IDs

Registration of VCF Operation is linked to a VCF Business Services console tenant and a Broadcom SiteID. The license allocations are linked to their contracts via serial numbers. If a contract or serial number is moved from one Broadcom SiteID to another, this impacts existing VCF Operations that are linked to the first SiteID with licenses from that SiteID.

Verify that you have a product or user administrator privileges for the new SiteID.

You cannot move serial numbers, you must contact support and request the move. In some cases, when you expand or renew a contract, if a new SiteID is added in the ordering process, serial numbers might be moved.

If you must move your serial number from one SiteID to another, you must first submit a usage report from VCF Operations and download a new license file. You must add licenses again in the new site. You must do this for all VCF Operations instances with a license from that serial number.

If one of your licenses appears as **Inactive** but your subscription is not expired, your serial number might have been moved.

When a serial number is moved, you might observe the following behavior:

1. Existing licenses linked to that serial number become **Inactive**.
2. Existing assignments appear as over assigned.
3. License updates fail.
4. After 180 days, your license expires, your hosts are disconnected from vCenter, and you cannot start workload operations.

After the serial number is moved, you must register your VCF Operations instance to the new SiteID and must re-license your environment.

1. Find the licenses and the VCF Operations instances that are related to the serial number.
   a) Log in to the VCF Business Services console.
   b) Navigate to **License Management** > **Licenses**.
   c) Use the search box to search for the serial number.

   All filtered licenses contain the searched serial number. The search results have a column for **VCF Operations Name.** These are the instances that must be re-linked to the new siteID.
2. Generate a license usage file from each VCF Operations instance that is related to the serial number, and update its licenses.
   a) Log in to the VCF Operations instance.
   b) Navigate to **Licensing** > **Registration**.
   c) Update licenses.

   - If the VCF Operations instance is in connected mode, verify that your license usage file was submitted within the last 24 hours by looking at the **Last Usage Report Generated** date, and click **Update Licenses**. For more information, see Update Licenses in Connected Mode.
   - If the VCF Operations instance is in disconnected mode, generate a license usage file. For more information, see Report License Usage and Update Licenses in Disconnected Mode.
3. After you move your serial number, for each VCF Operations instance, create your licenses again by splitting the default license. For more information, see Managing Licenses in the VCF Business Services Console.

   As a best practice, allocate the same capacity to the licenses as the capacity from the prior SiteID. Consider also using a distinct license name from the prior site.
4. Deactivate the VCF Operations instance in the VCF Business Services console. For more information, see Deactivate a VCF Operations Instance.

   You must select the check box **I no longer have access to the VCF Operations instance**.
5. Restart registration for each VCF Operations instance that is linked to the serial number.
   a) Log in to the VCF Operations instance.
   b) Navigate to **Licensing** > **Registration**.
   c) Click **Actions > Restart Registration.**

   Important:

   This action does not impact the existing licenses in your environment.
6. Register each VCF Operations instance linked to the serial number with the new SiteID. For more information, see Registering VCF Operations and a License Server with the VCF Business Services Console.

   CAUTION:

   During registration, select the new Site ID, and the license you created in step 3.
7. Relicense your environment by assigning the licenses to the vCenter instances. For more information, see Assign a Primary License to a vCenter Instance.

   Both your old and new licenses are displayed in VCF Operations. After your next license update, if the old licenses are no longer assigned to vCenter instances, they do not appear in VCF Operations.

Update your licenses. For more information, see Updating Licenses and Viewing the License Usage File.

A License Server Does Not Appear in the License Servers Table in VCF Operations

A license server does not appear in the list of license servers in VCF Operations but appears in the VCF Business Services console.

If you take a snapshot of VCF Operations before adding a license server to it, when you revert to that snapshot, the license server is not available in the VCF Operations instance but remains in the VCF Business Services console.

You cannot reconnect the license server to the VCF Operations instance. You can transfer the licenses from that license server to a new license server that is connected to the VCF Operations instance, and delete the old license server.

Note:

After you upgrade VCF Operations from version 9.0 to version 9.1, a license server is automatically deployed and must register it with the VCF Business Services console. After the upgrade, even if you revert to a snapshot of VCF Operations before the upgrade, you cannot roll back to version 9.0 in the VCF Business Services console. License-related operations such as license usage reporting might fail.

1. Optional: Transfer the licenses from the license server to a license server that is connected to VCF Operations.

   The licenses are automatically assigned to the assets that they were assigned to before the transfer of licenses.
2. Delete the license server from the VCF Business Service console. For more information, see Delete a License Server.
3. Delete the license server virtual machine from the vCenter instance.